PHYS 7501:
Particles and Nuclei I

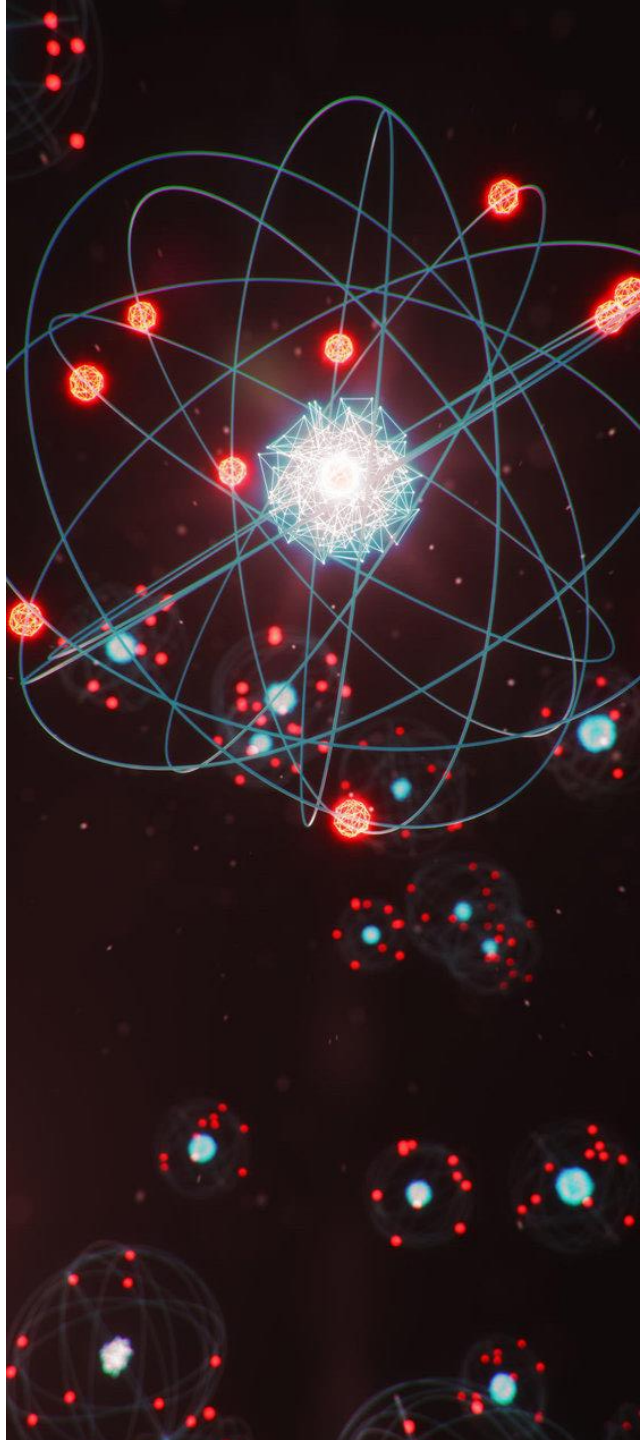
Tues./Thurs. 2 – 3:20PM
Clippinger 105

Dr. Andrea Richard

Edwards Accelerator Lab 204

richarda1@ohio.edu

Available via e-mail & Teams

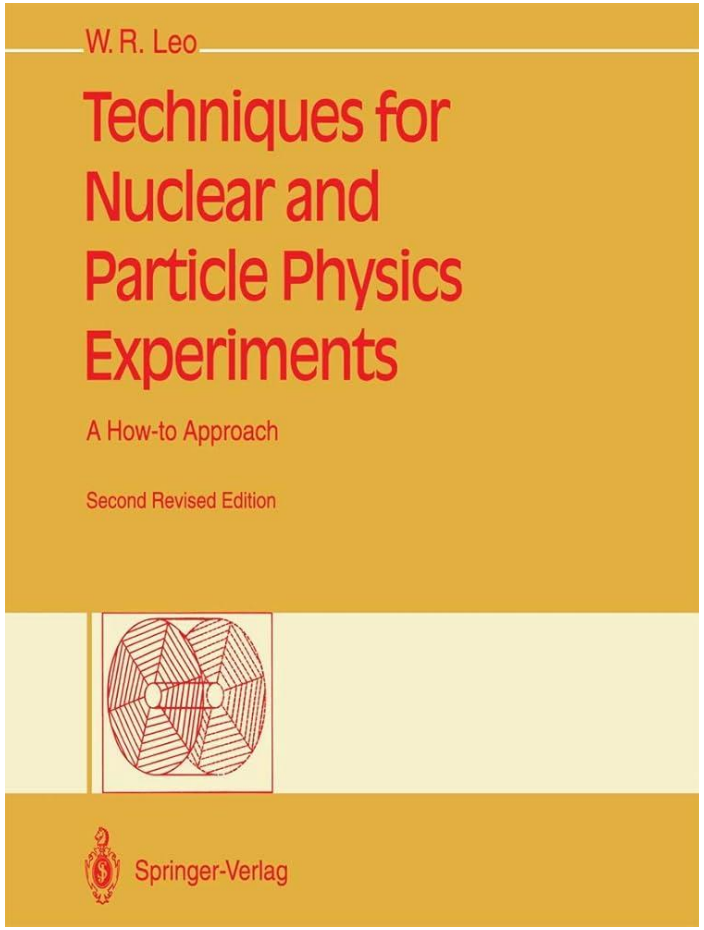


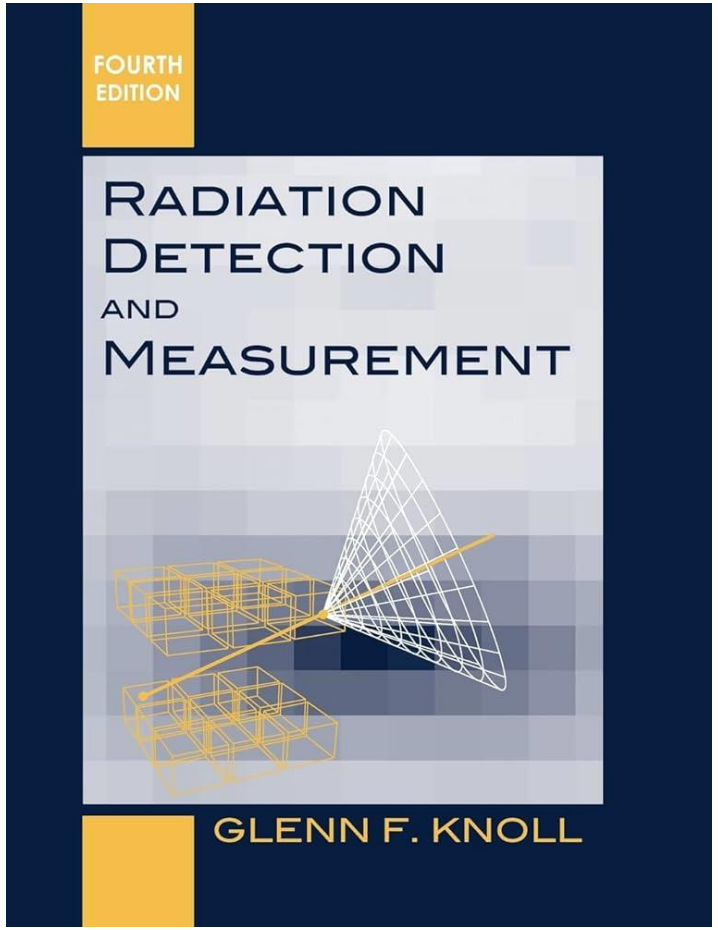
EXPERIMENTAL TECHNIQUES

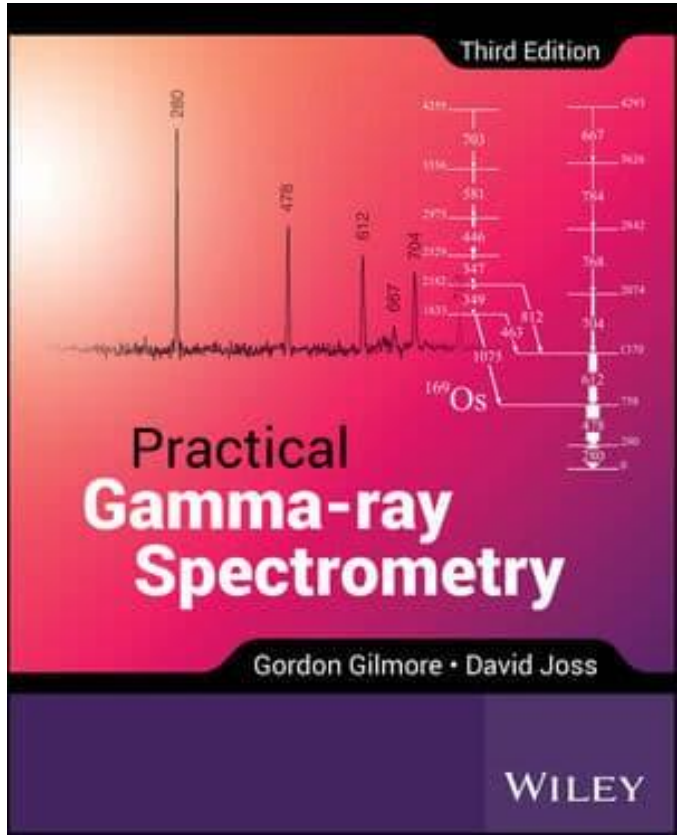
Modern Nuclear Chemistry, Chapter 13, 17, 18

Monday	Tuesday	Wednesday	Thursday	Friday
11/17	11/18	11/19	11/20	11/21
	Experimental Techniques **Paper Outline Due**		**Nuclear Medicine**	
11/24	11/25	11/26	11/27	11/28
	Nuclear Energy **Draft of Slides Due**		Thanksgiving Break	
12/1	12/2	12/3	12/4	12/5
	TBD		**First Day of Final Presentations**	

Resources for Experimental Nuclear Science

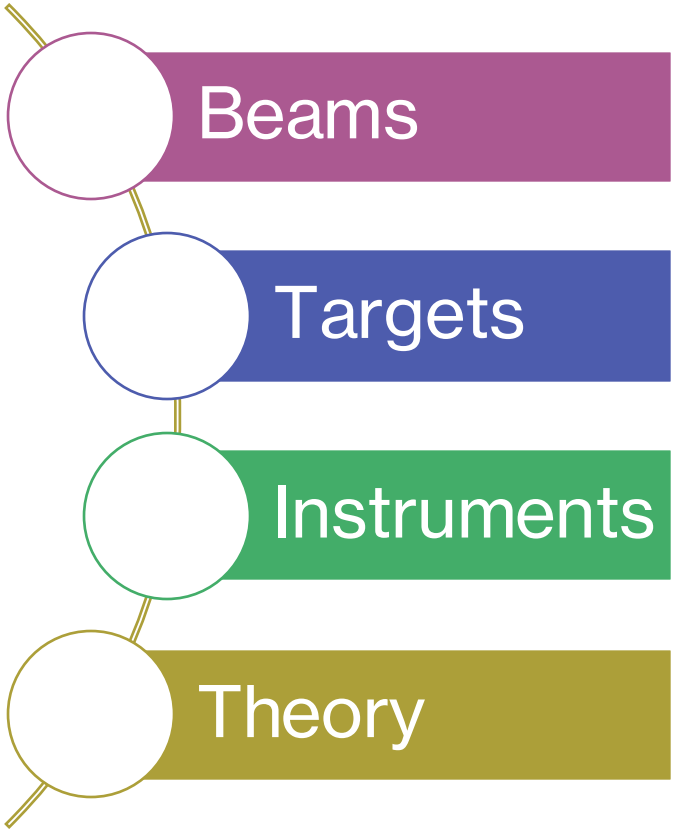






“Ingredients” of an experiment

- **Accelerator to provide the beam (reactors too)**
  - stable/radioactive, pure/cocktail, intensity
  - energies from keV to 100s MeV/A
- **A target**
  - solid/liquid/gaseous, density/thickness
  - active targets
- **Instruments**
  - HPGe detectors
  - Double-sided strip detectors (Si,Ge)
  - scintillators- magnetic spectrometers
  - gaseous detectors
  - ion traps
  - ...
- **Theory**
  - interpretation of the data
  - motivate experiments to validate theories



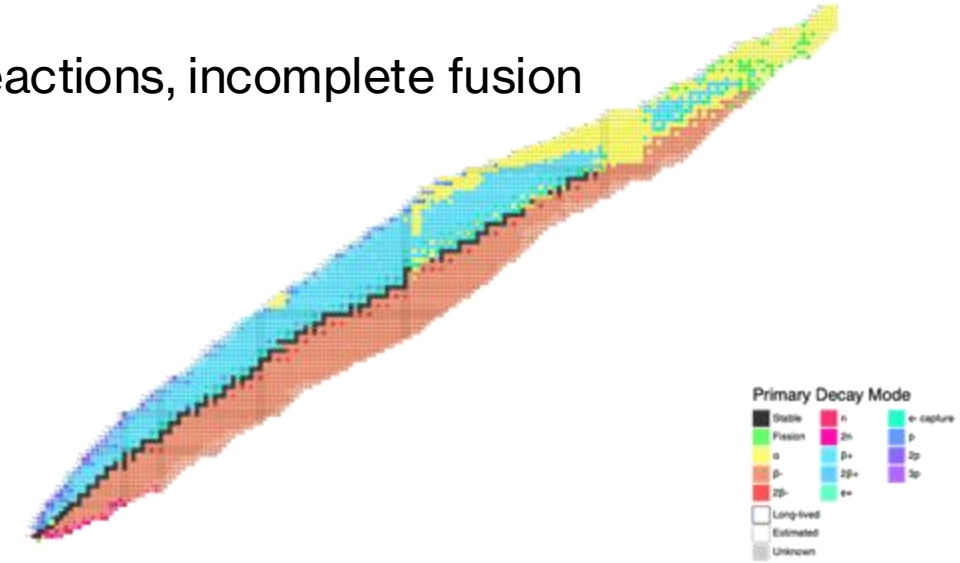
Production of Beams

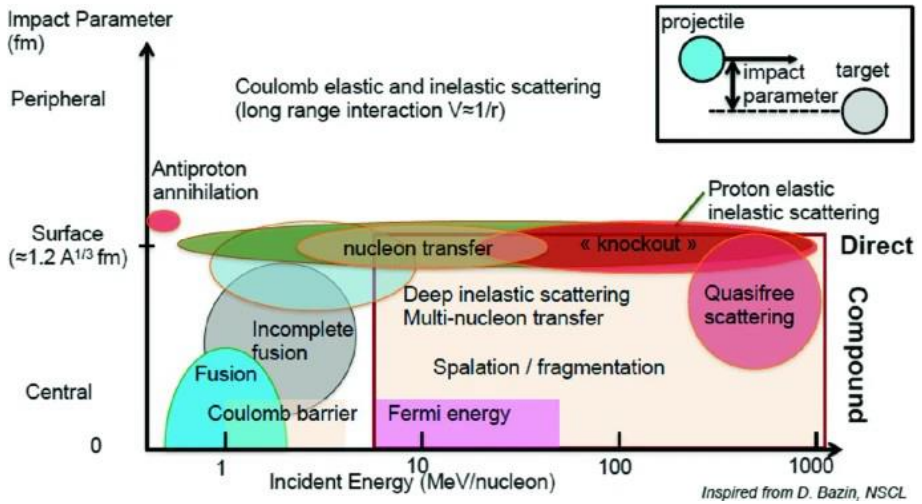
Reactions

- Coulomb excitation (low and high energy)
- (deep-)inelastic scattering, (multi-nucleon) transfer reactions, incomplete fusion
- fusion-evaporation
- target/beam fragmentation
- nucleon knockout

Decay

- alpha-decay
- beta-decay
- fission (spontaneous/induced)





Where can you get stable beams?

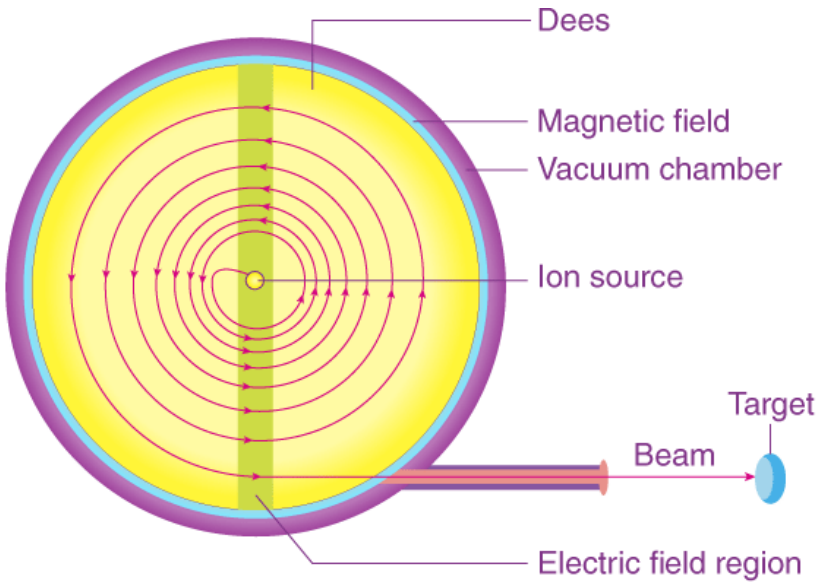
Association for Research at University Nuclear Accelerators

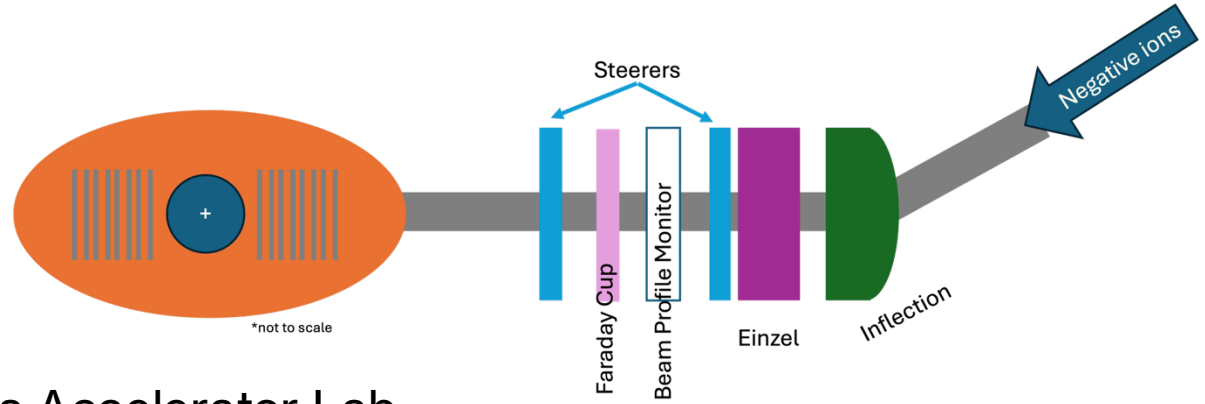


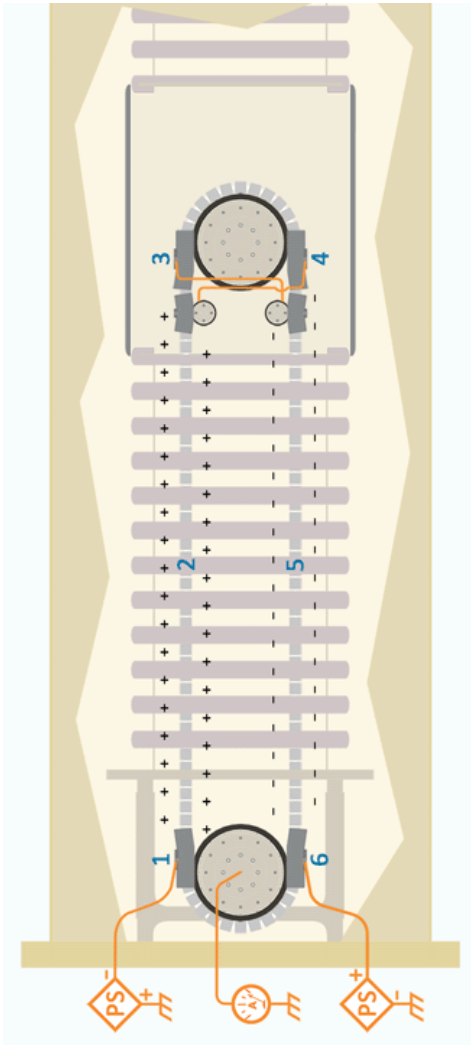
Plus others like ANL's ATLAS facility, iThemba in South Africa, Oslo Cyclotron Lab in Norway, ++

Stable Beam Production

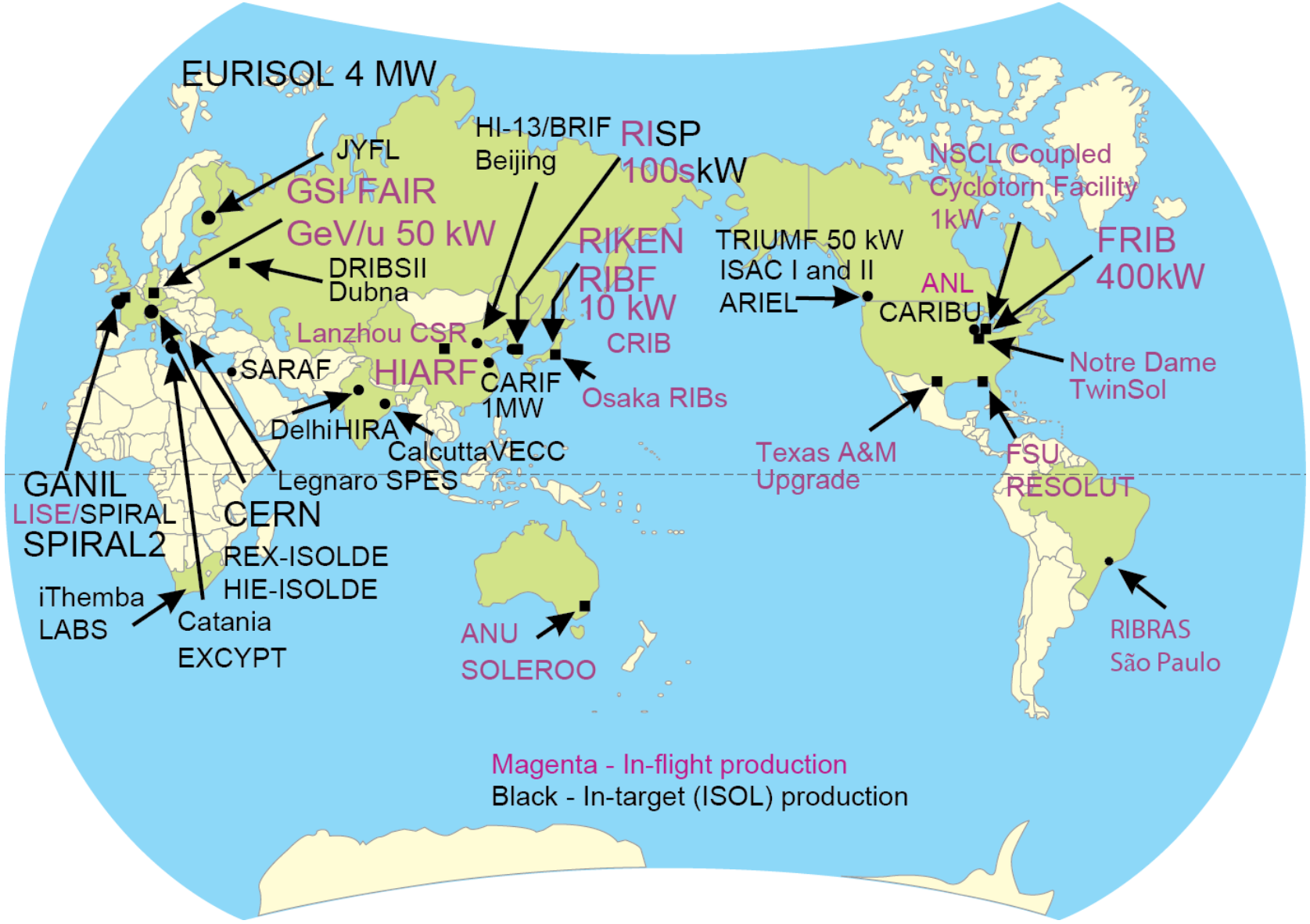
- Ion sources produce the beams
- Many varieties of accelerators: LINAC, Tandem Van deGraaff, Cyclotron





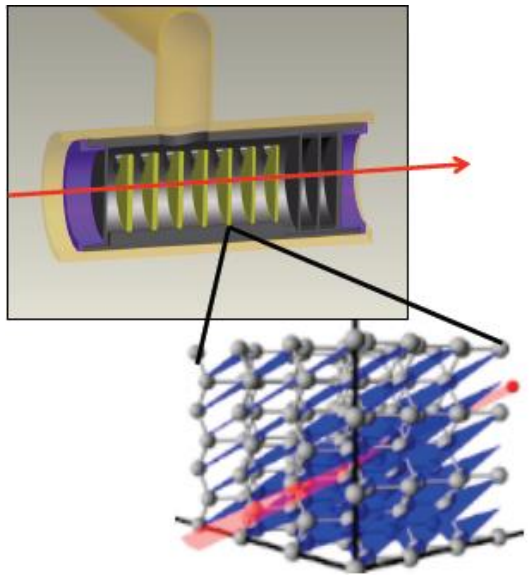


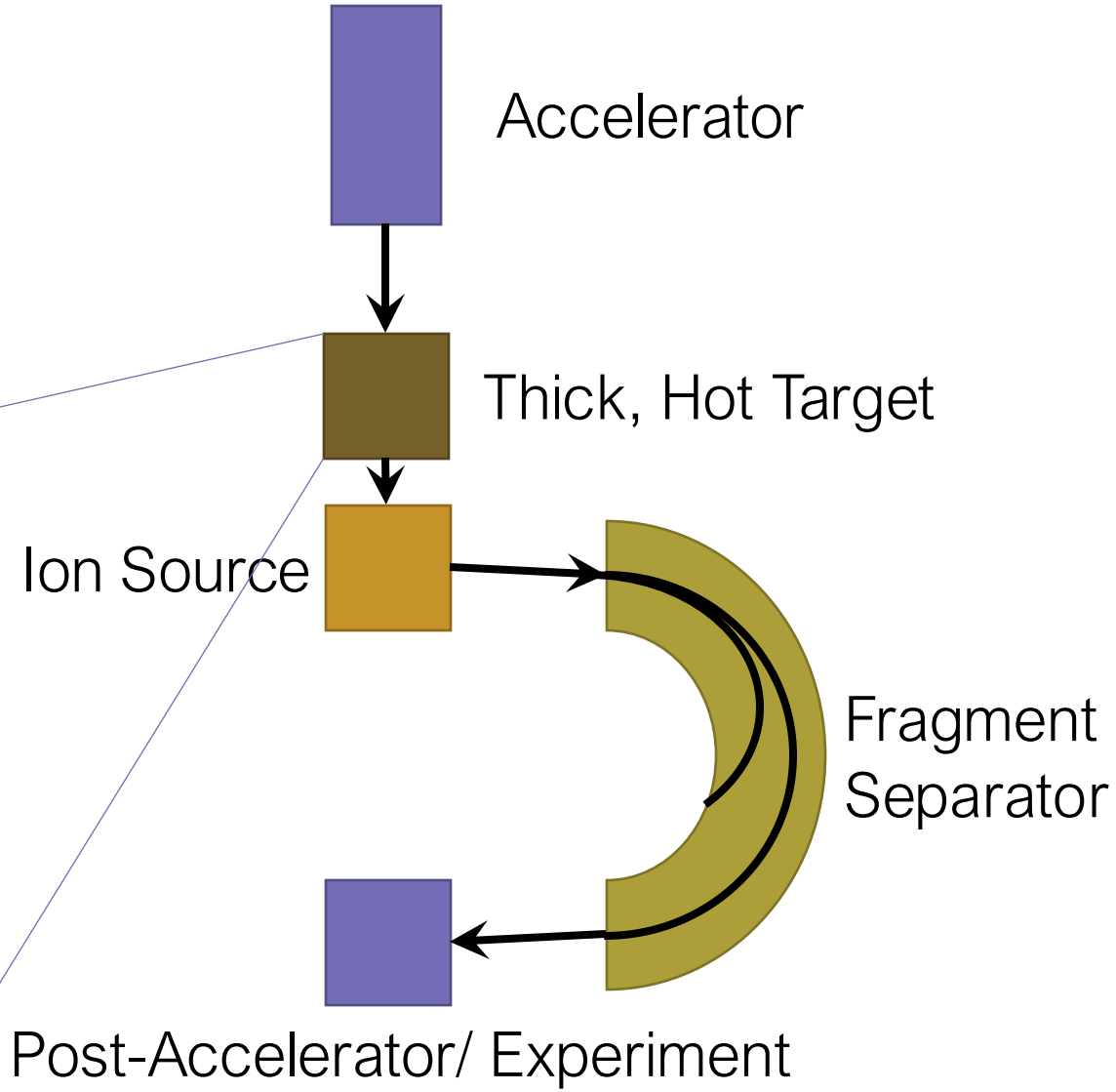
Where can you get radioactive ion beams?



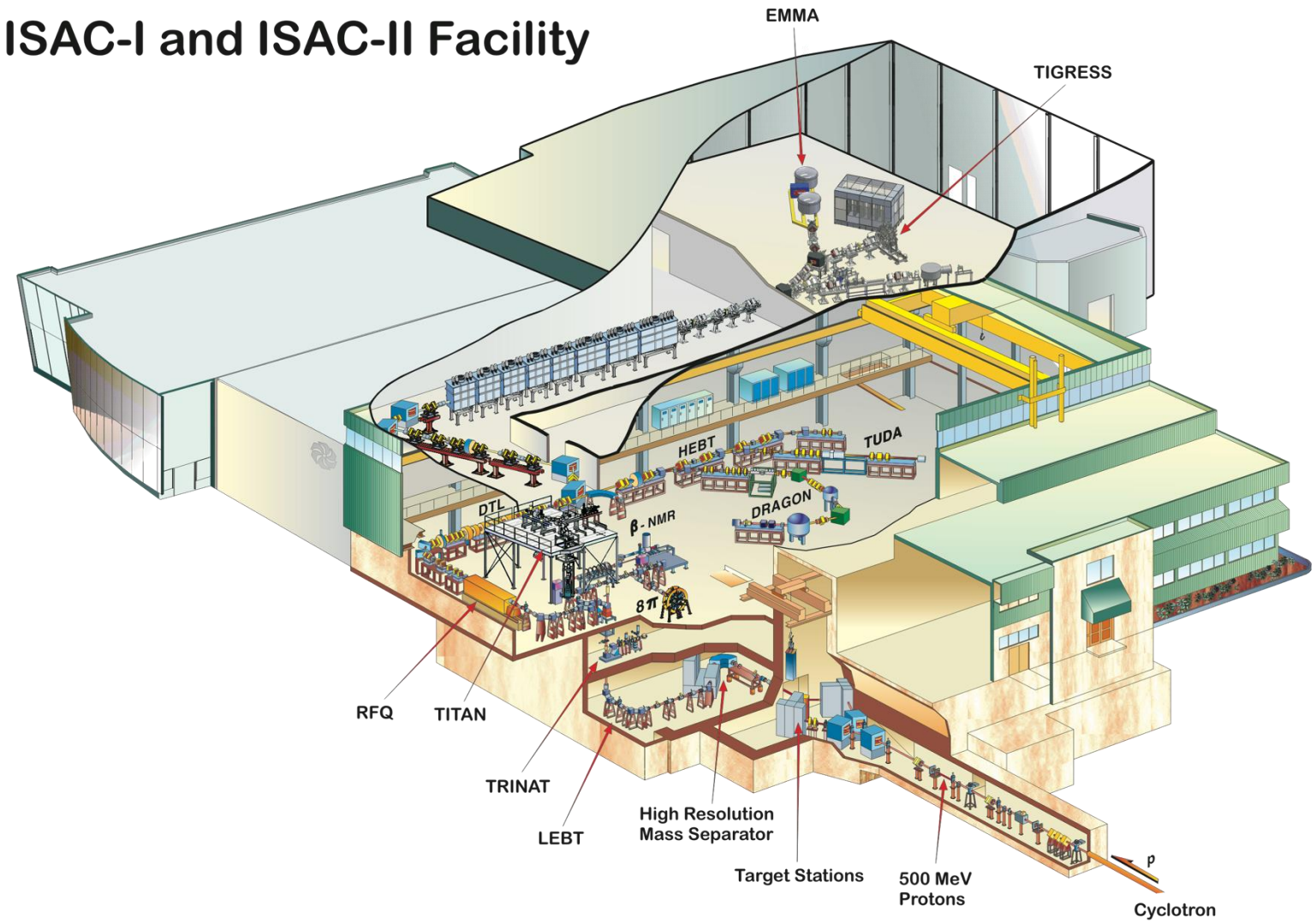
Isotope Separation On-Line (ISOL)

Examples of ISOL facilities:
TRIUMF (Canada)
HRIBF (USA)
SPIRAL/SPIRAL2 (France)
REX-ISOLDE/HIE-ISOLDE (CERN)
iTHEMBA - future radioactive beam facility (South Africa)



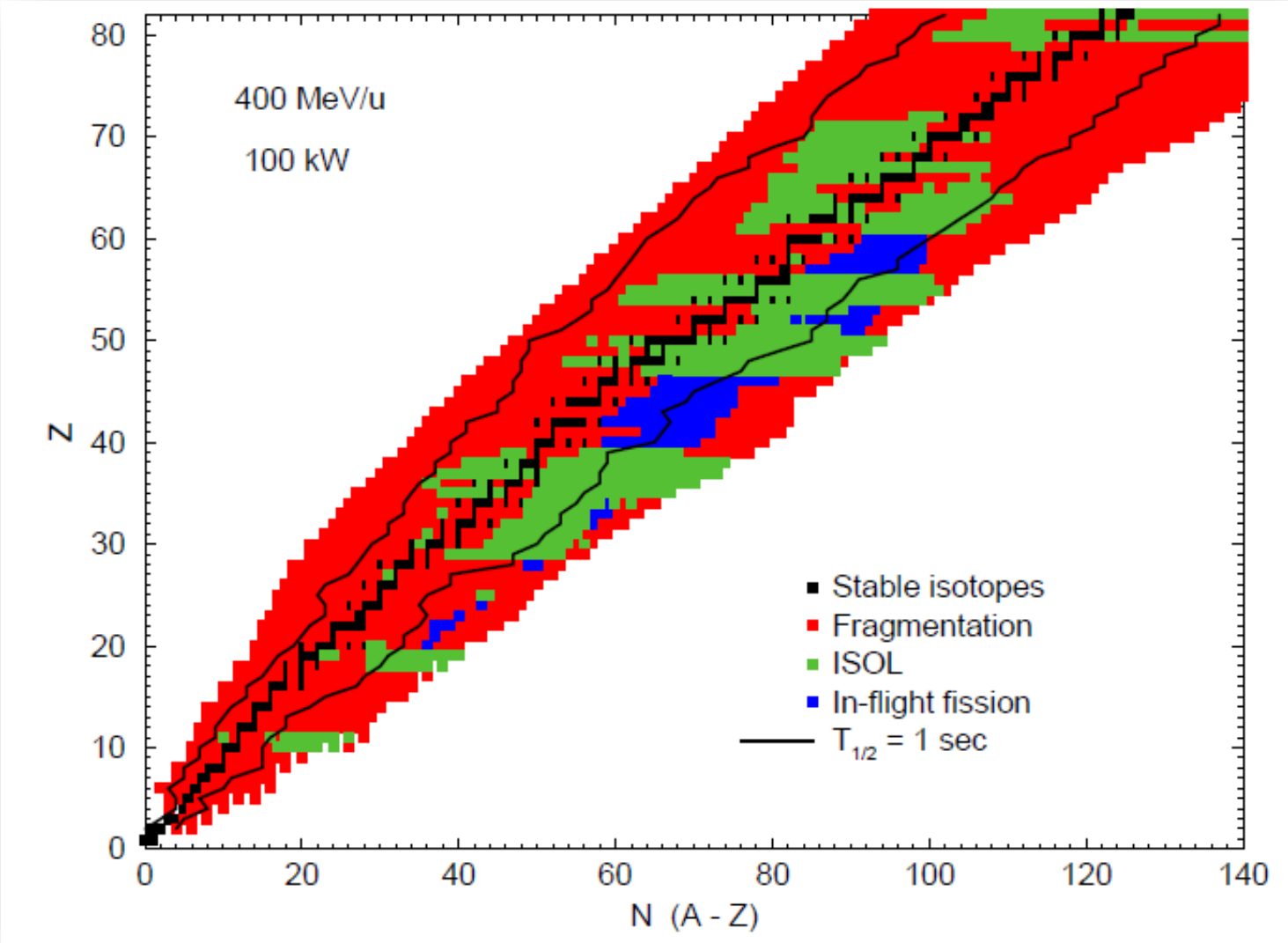


ISOL Facilities - TRIUMF

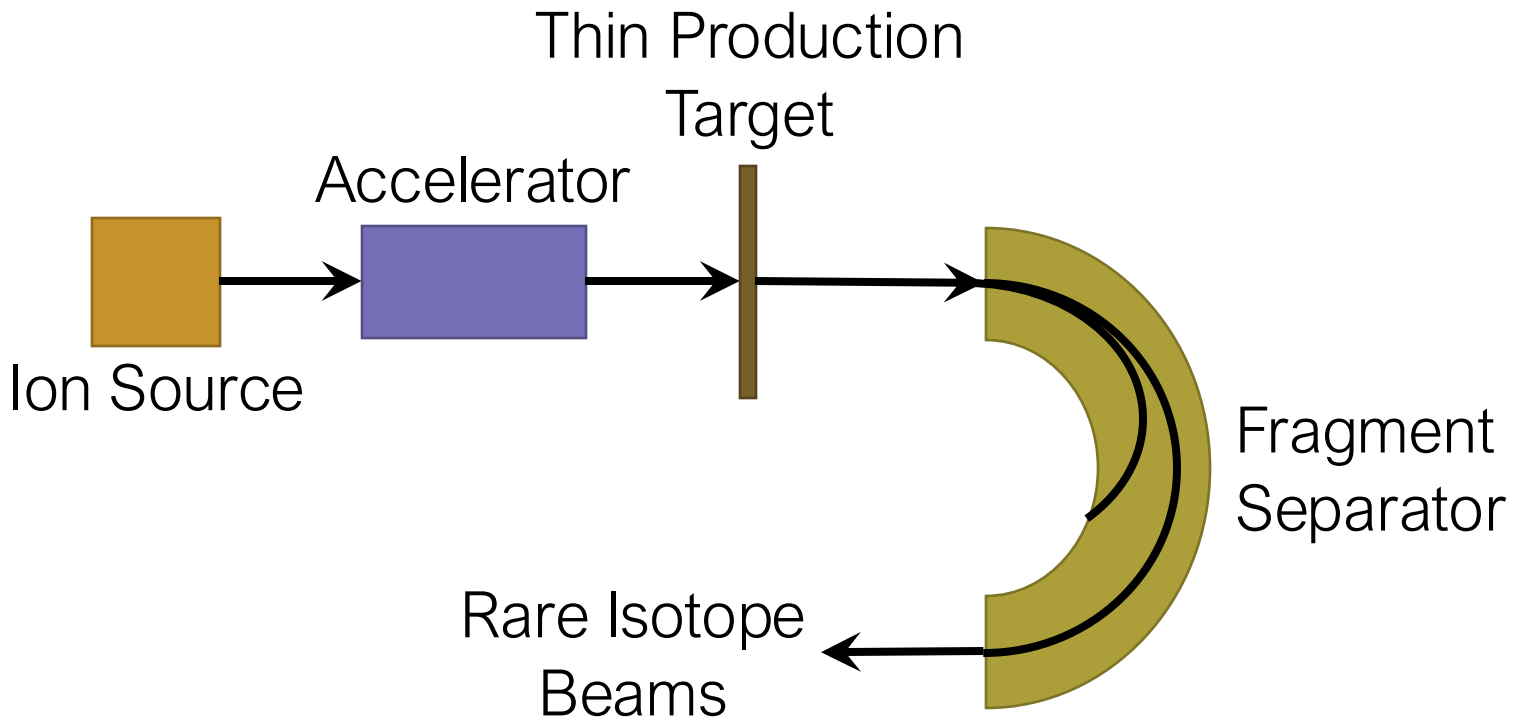


- TRIUMF is home to the world's largest cyclotron – accelerates H- to 520 MeV, extracts as proton beam
- Proton beam is sent into target hall, and interacts with thick targets (materials such as UC)

What can we make? – ISOL

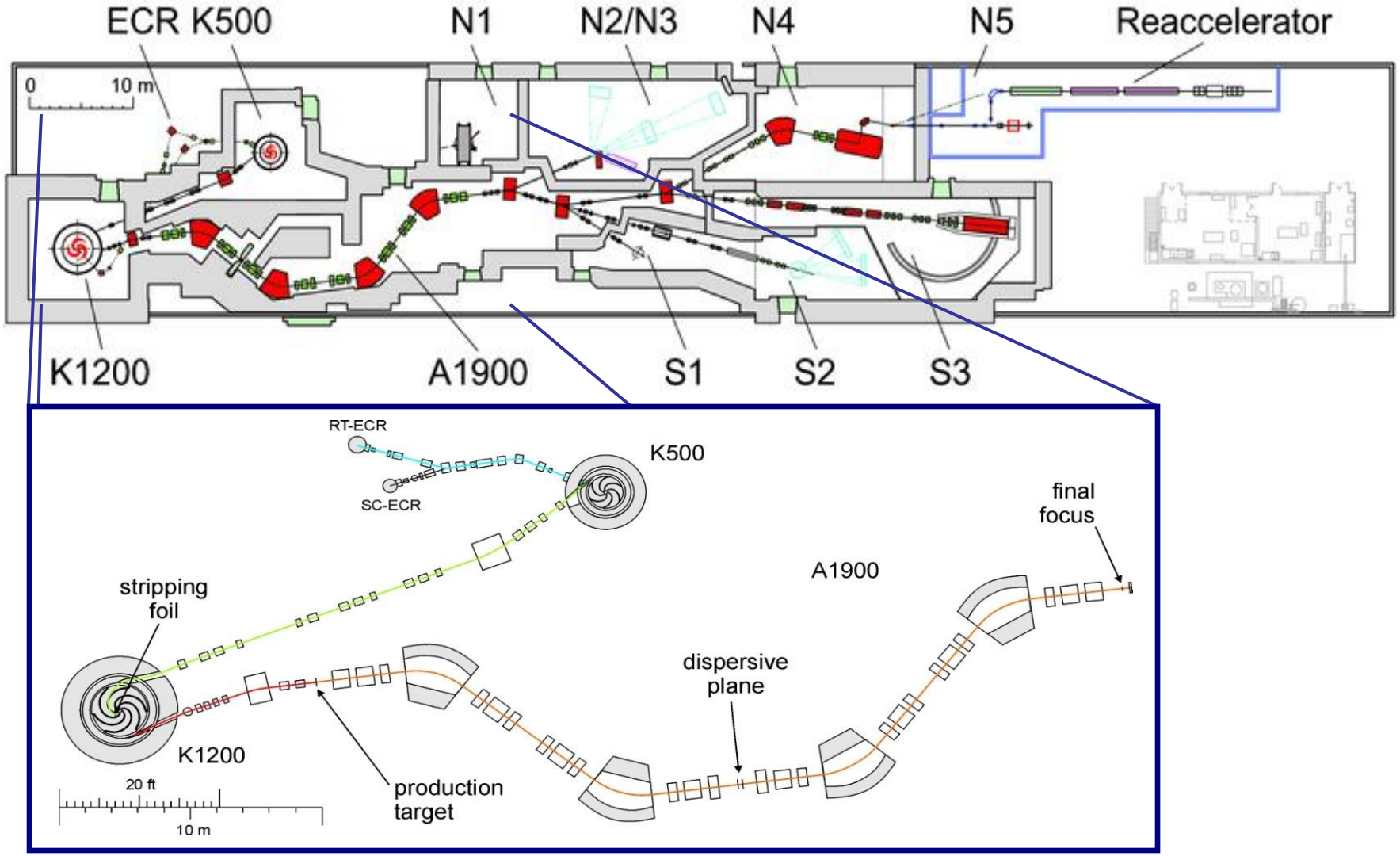


Fragmentation + In-flight separation

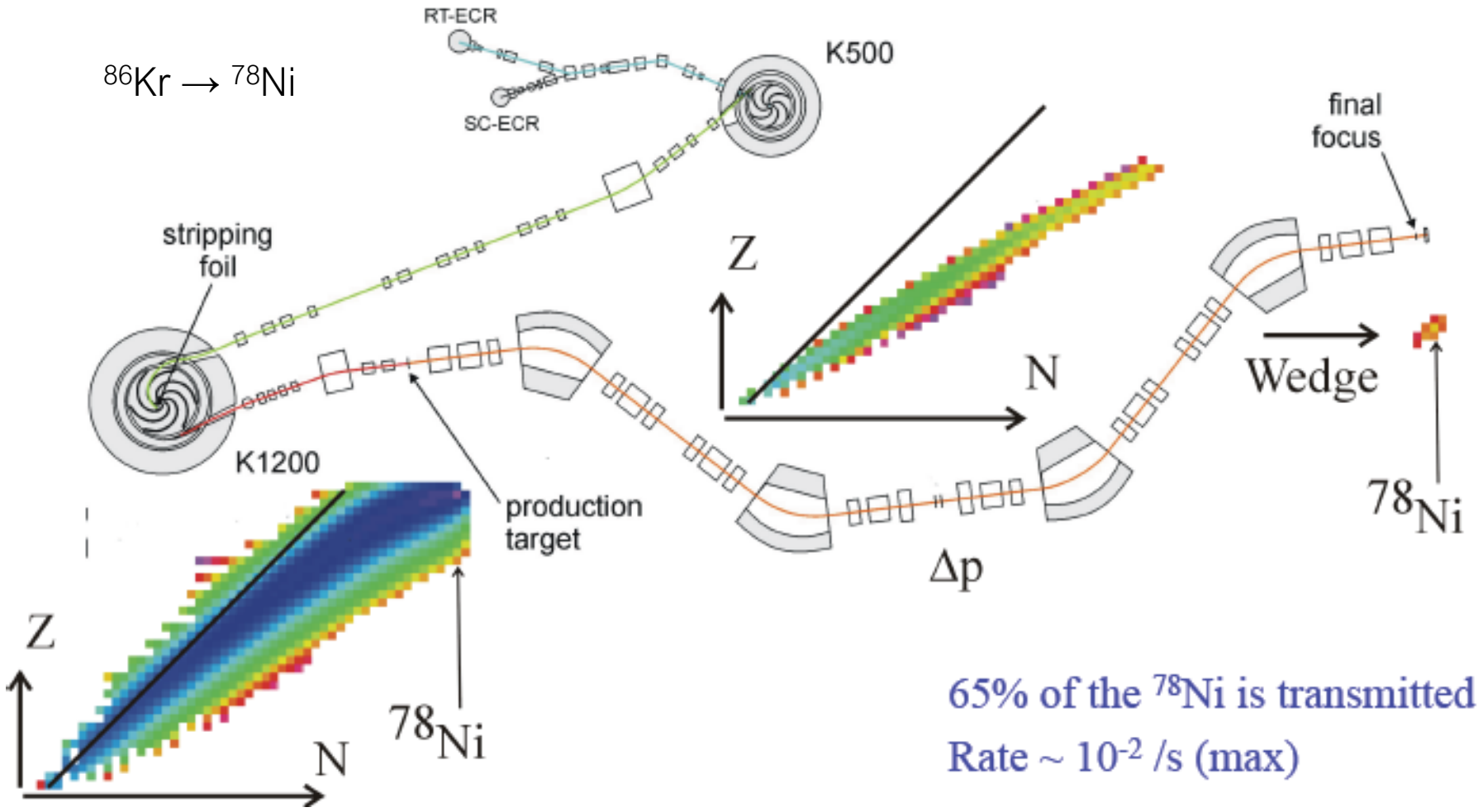


Examples of fragmentation facilities:
NSCL (USA) --> FRIB
RIKEN (Japan)
GANIL (France)
GSI (Germany)

Fragmentation facilities - NSCL



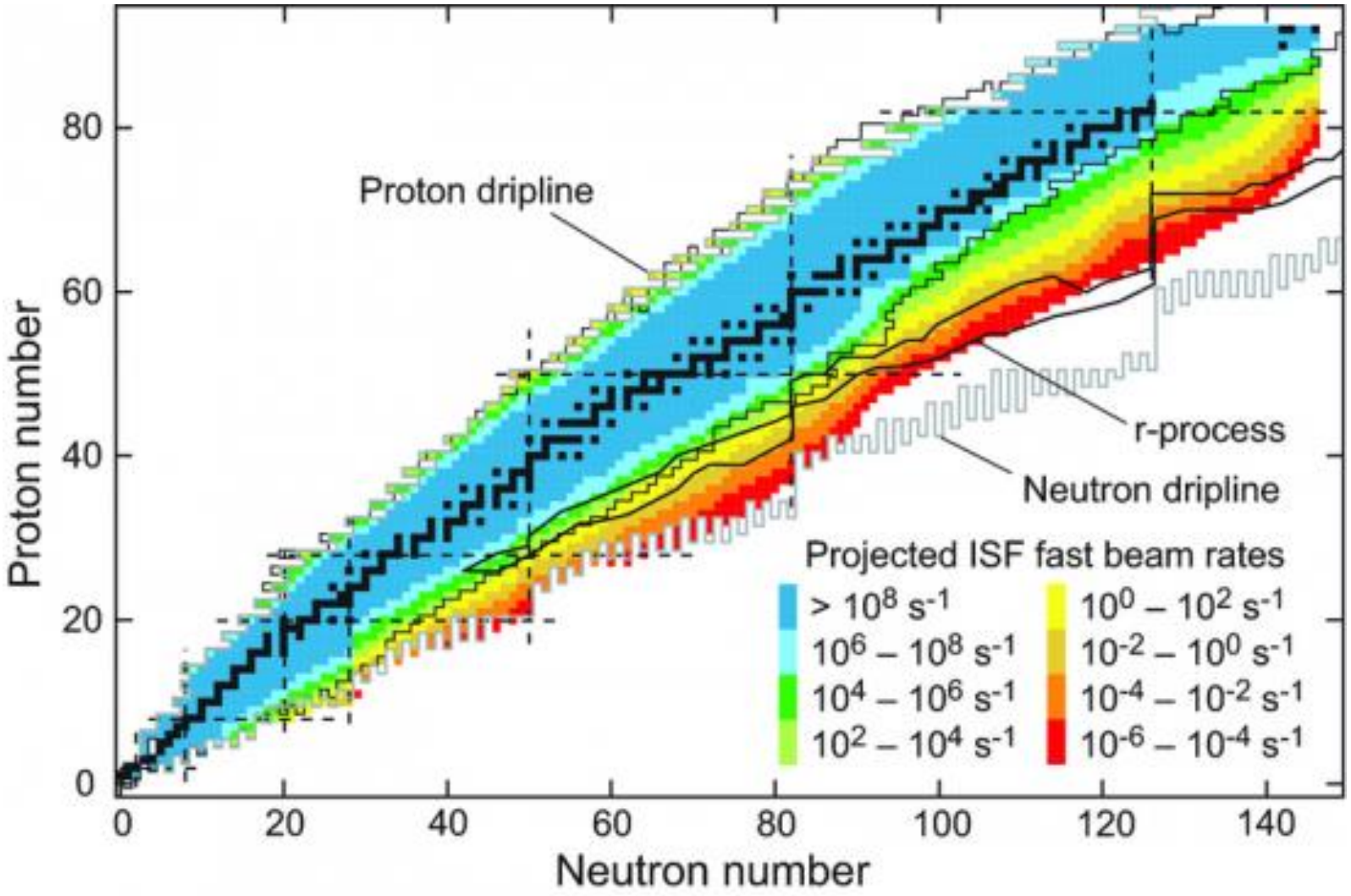
Fragmentation + separation: Example



65% of the ^{78}Ni is transmitted

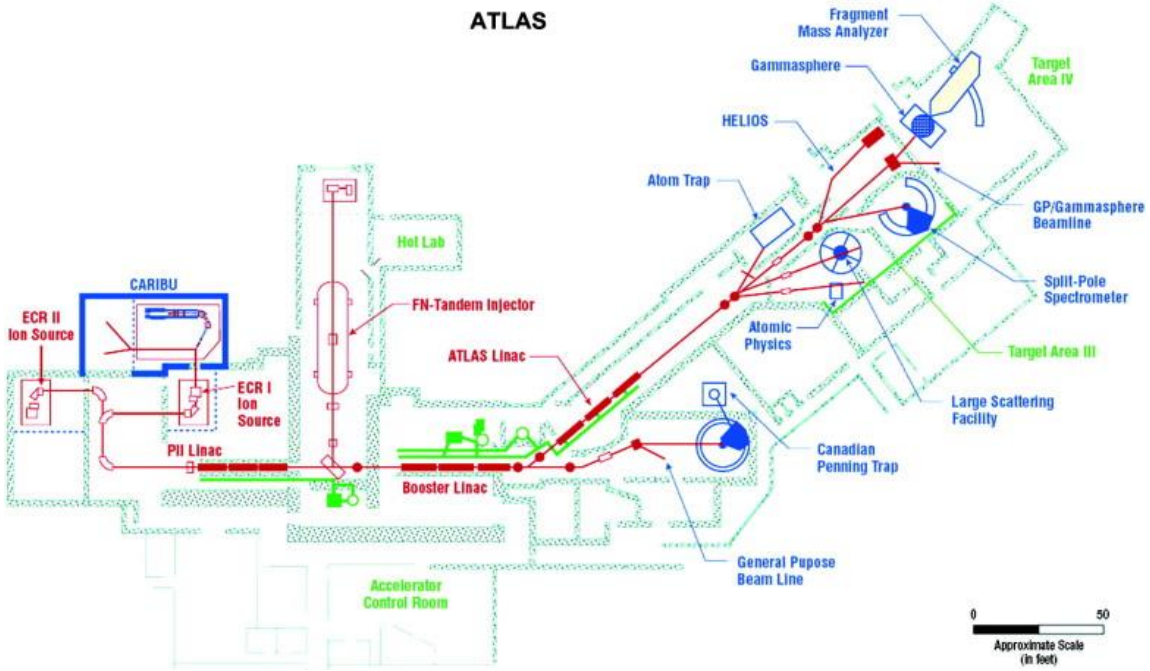
Rate ~ 10^{-2} /s (max)

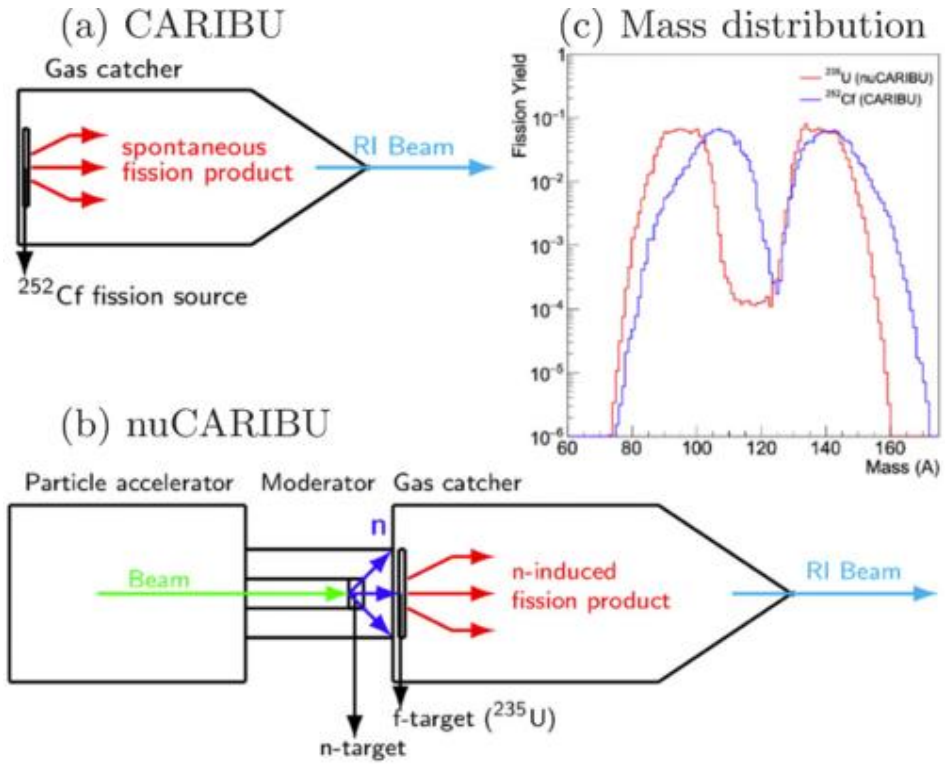
What can we make? – Fragmentation



Fission-fragment accelerator

- Turns a source of neutron-rich isotopes, such as a spontaneous fission source (CARIBU) or neutron-induced fission (nuCARIBU), into a low-energy beam using a gas catcher and charge breeder





https://doi.org/10.1103/PhysRevC.109.024605

Fission fragment yields

Production has inherent selectivity – cannot produce all nuclei but will provide an intense source of certain species.

Due to reacceleration, beams should, like in ISOL facilities be of very high quality.

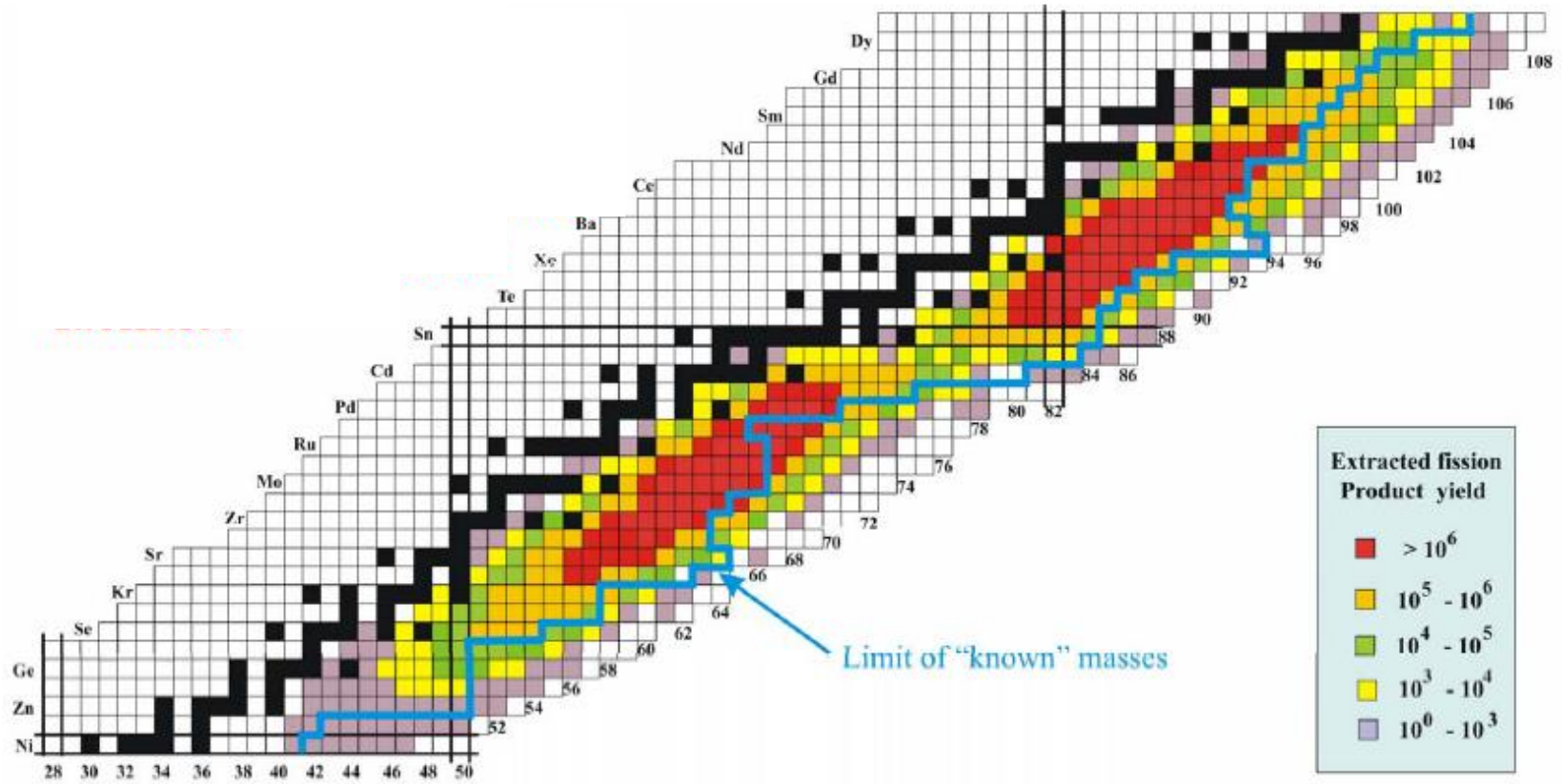
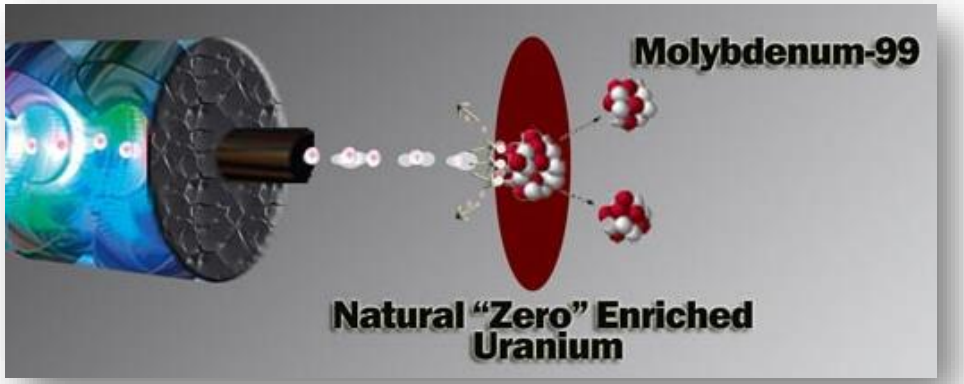


Figure: CARIBU Upgrade presentation @ ATLAS User Group Workshop, 2004

On the horizon: Photo-fission production



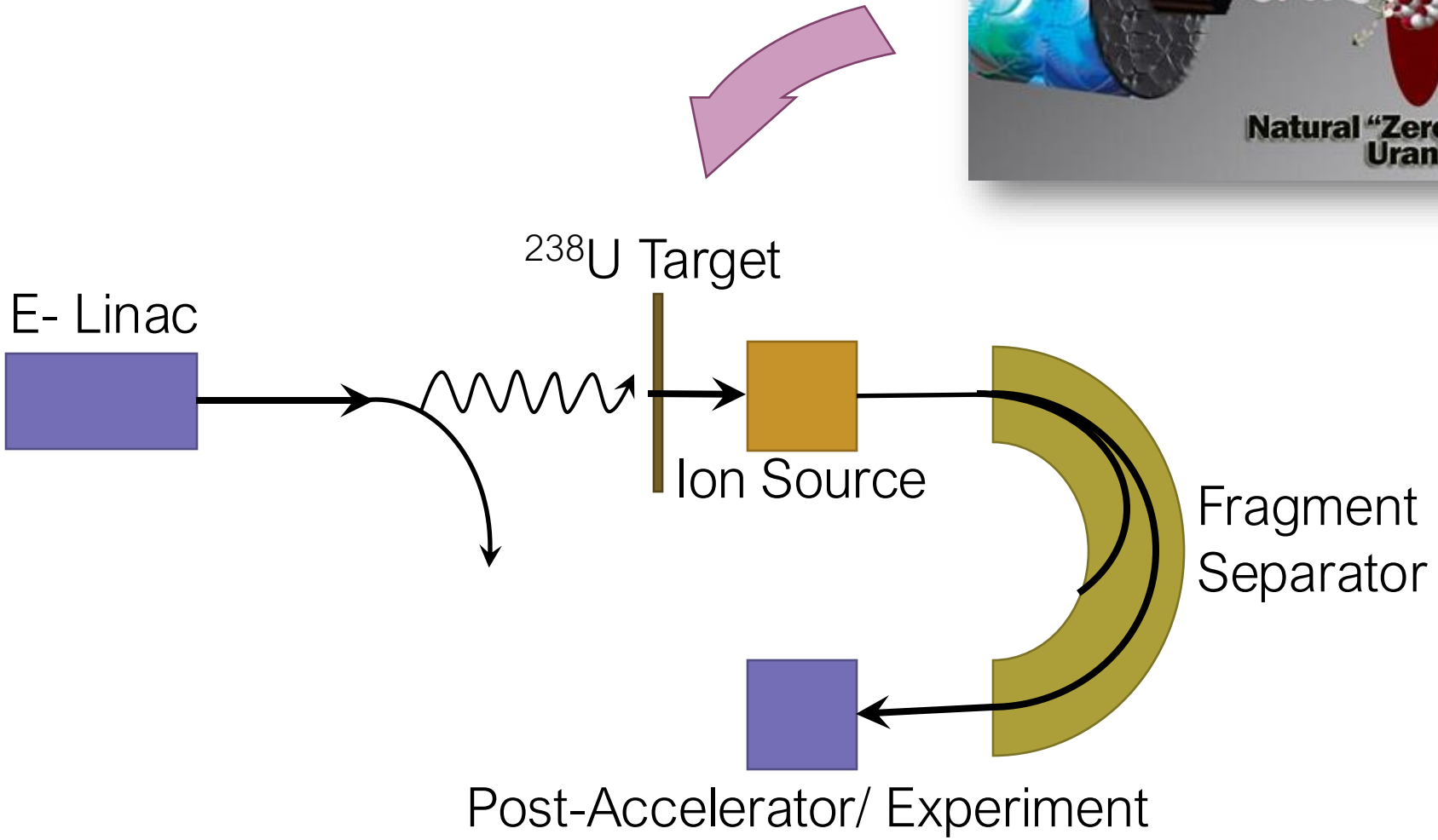
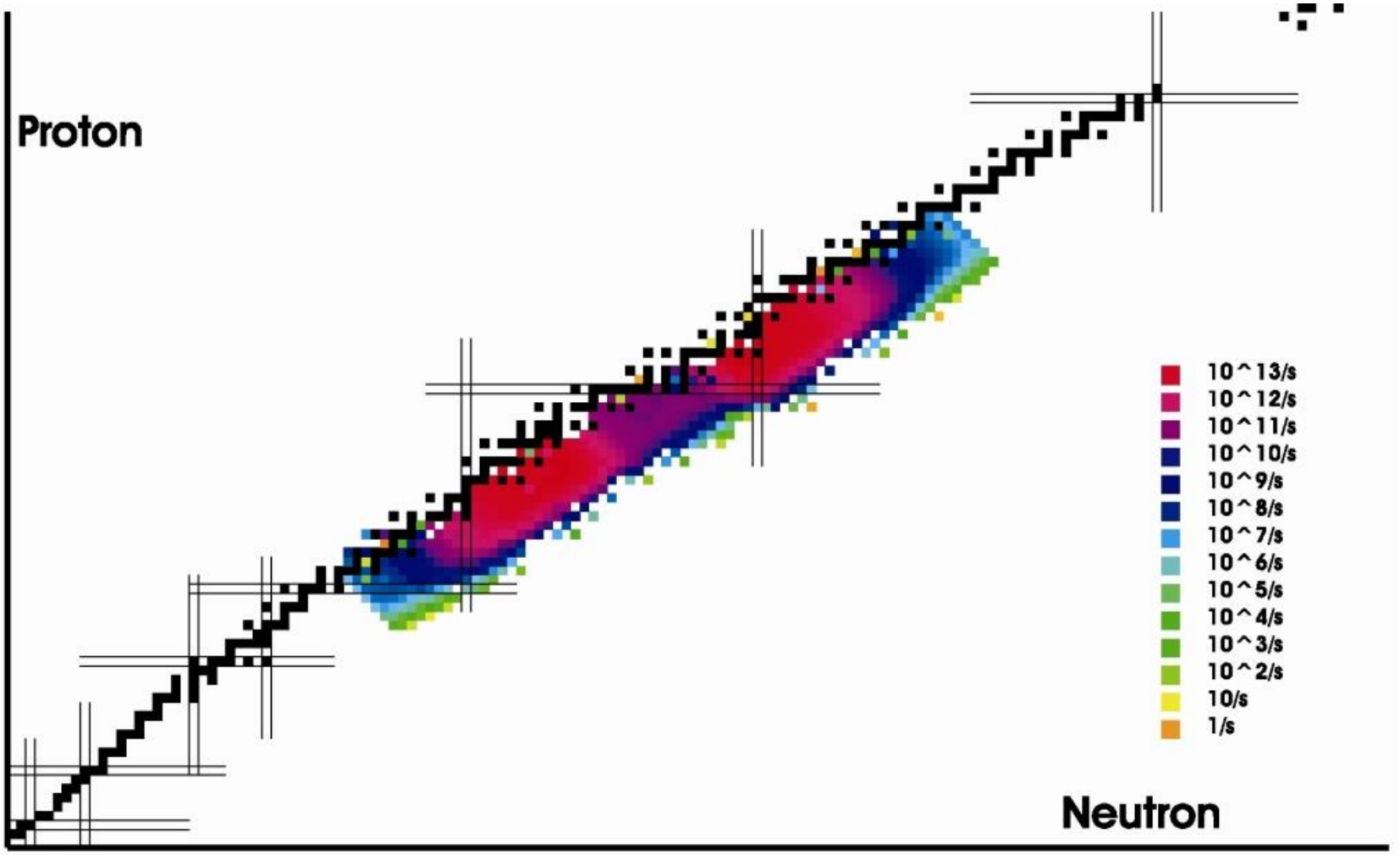


Photo-fission facility: ARIEL @ TRIUMF



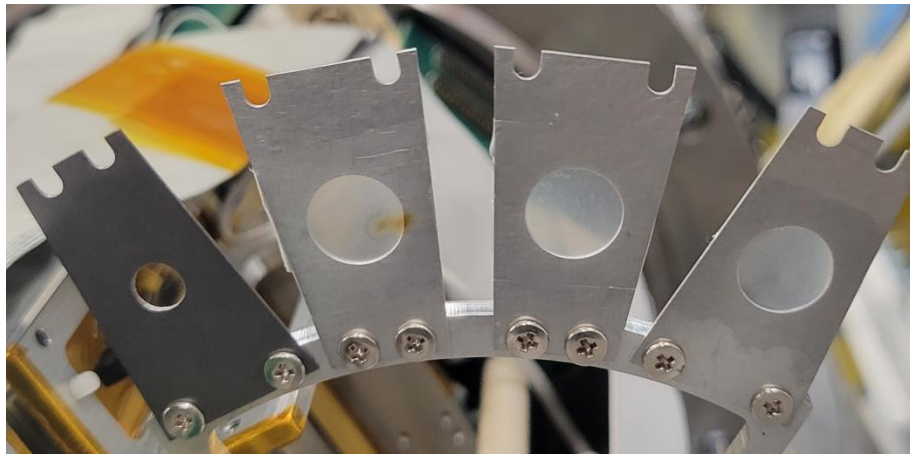
ARIEL: www.triumf.ca/ariel

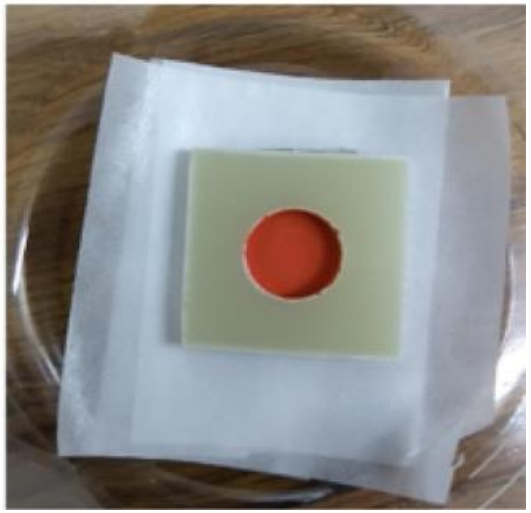
Targets



JENSA Gas Target, ORNL @ FRIB
Jet Experiments for Nuclear Structure and Astrophysics

CD_2 targets, one slightly burned by beam





Solid Mercury targets

https://doi.org/10.1016/j.nima.2022.167087



^{100}Mo targets

Instruments

How do we detect all of these reaction products and decays that we've been discussing?

- We use different types of detectors for different particles.

How do we process signals from detectors?

- Data Acquisition Systems (Digital and/or Analog)

General Characteristics of Detectors

- **What we want to measure:**
  - → energy, velocity, momentum, mass
  - → time
  - → position
  - → cross section
  - → correlations (angular, feeding)
- **We then want to infer (with theory):**
  - → spin/parity
  - → angular momentum transfer
  - → spectroscopic factors
- **What we actually measure:**
  - → charge or light (pulse)
  - → time (from a defined clock)
  - → segment ID (pixel hit)
  - → number
  - → true/false logic gates
- **Things we can optimize:**
  - → resolution
  - → efficiency
  - → rate capability
  - → selectivity

General Characteristics of Detectors

- We need a device that will convert incoming radiation into a signal that we can measure
- Generally, there are three classes of detectors
  - **Semi-conductor detectors**: ionization (electron-hole pairs) collected into current or charge signal
  - **Scintillation detectors**: deexcitation (photons), converted to electrons via a photocathode/photodiode and collected into current or charge signal
  - **Gas-filled detectors**: ionization (electron-ion pairs) collected into current or charge signal

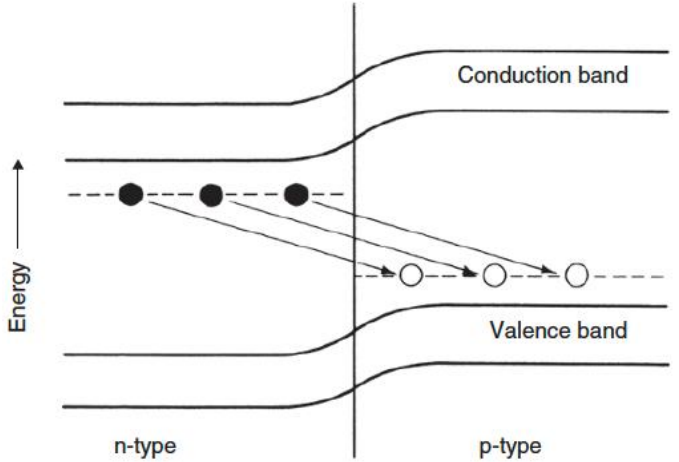


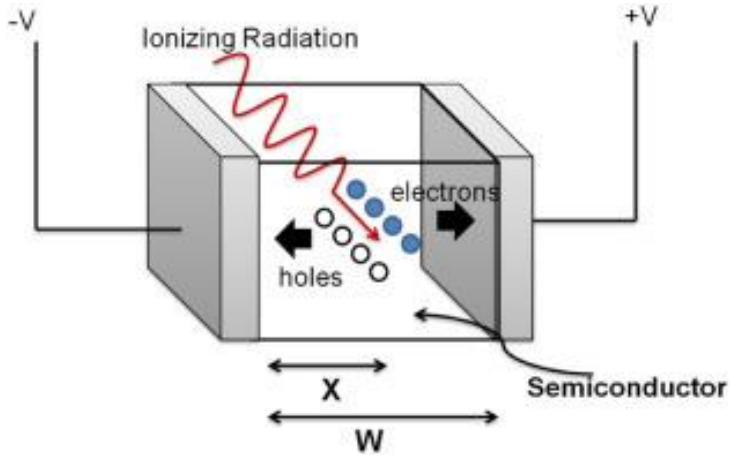




Semi-Conductor Detectors

- Also known as “solid-state” detectors
- Silicon, Germanium = semi-conductors
- “intrinsic” semiconductors are rare and hard to produce in large quantities, so we want to dope the semiconductor to tailor its response to our needs
- *n-type*: donor electrons from doping (pentavalent dopant in the case of silicon), lie very close to conduction band, net effect is to create far more conduction electrons than holes
- *p-type*: donor holes (or “acceptors”) from doping (trivalent dopant in the case of silicon), lie very close to valence band, net effect is to create more conduction holes as the charge carrier
- incoming radiation creates electron-hole pairs along track (roughly 3 eV per pair), these charge carriers are migrated through the material with an applied electric field and the current signal, proportional to energy, is read out from electrical contacts





Scintillation Detectors

- A scintillation detector consists of a scintillating material, coupled to a light guide, and a photo detector.
- Two types: organic and inorganic
  - Differ by how light is produced
- **Inorganic scintillators**: NaI (Tl), CsI (Tl), BGO – $Bi_4Ge_3O_{12}$, BaF_2, ...
  - Scintillation properties are different depending on the crystal
  - Scintillation mechanism is usually due to the band structure of the crystal (band gaps ~ 5-10 eV)
- **Organic scintillators**: plastics (e.g. benzene C_6H_6, stilbene, naphtalen, NE-213)
  - Scintillation mechanism is due to the transition of electrons between molecular orbitals

Scintillation Detectors

- Regardless of organic vs. inorganic, light pulses need to be converted into measurable signals.
- This is done using a photoelectric material (photo-cathode) to generate electrons from incident photons
- Because just a few photons are “detectable,” the signals are small, so we need electron multiplication (gain)

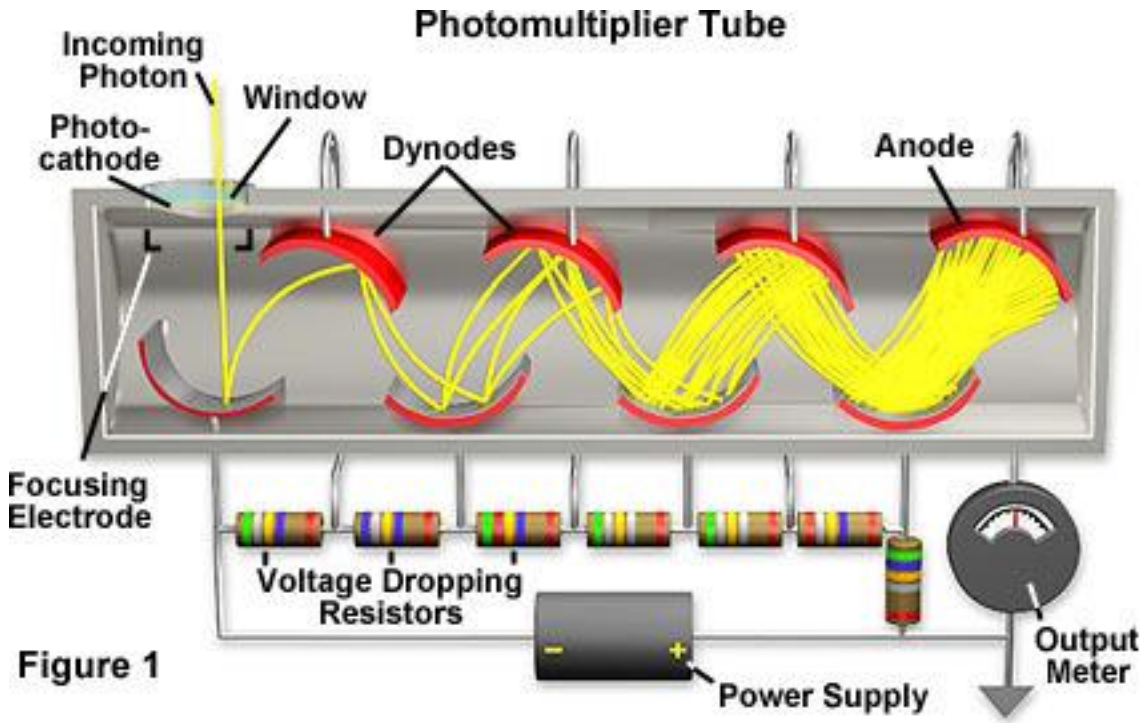


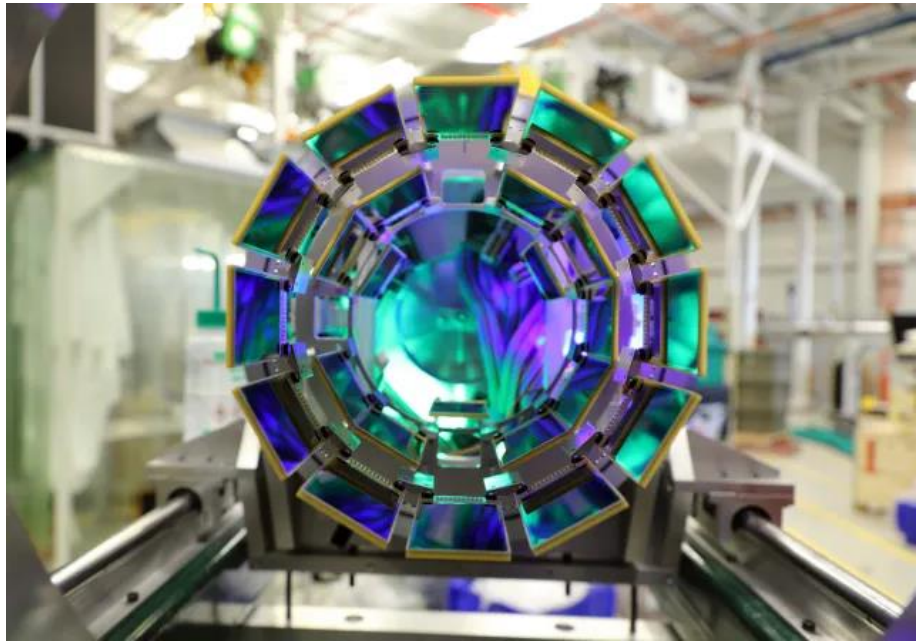
Figure 1



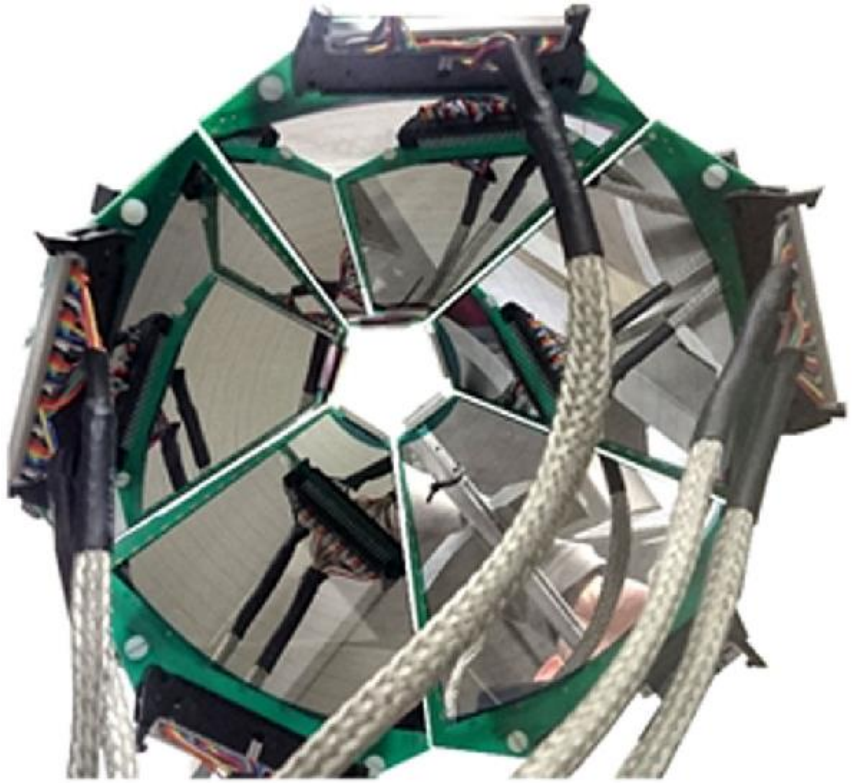
Charged Particle Detection

- Mostly semi-conductors and usually silicon, but other examples exist.
- Typically very high detection efficiency





ORRUBA



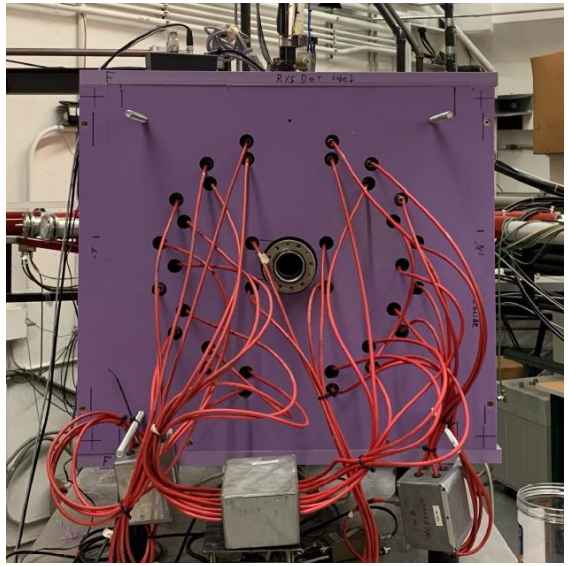
CAKE @ iThemba

Neutron Detection

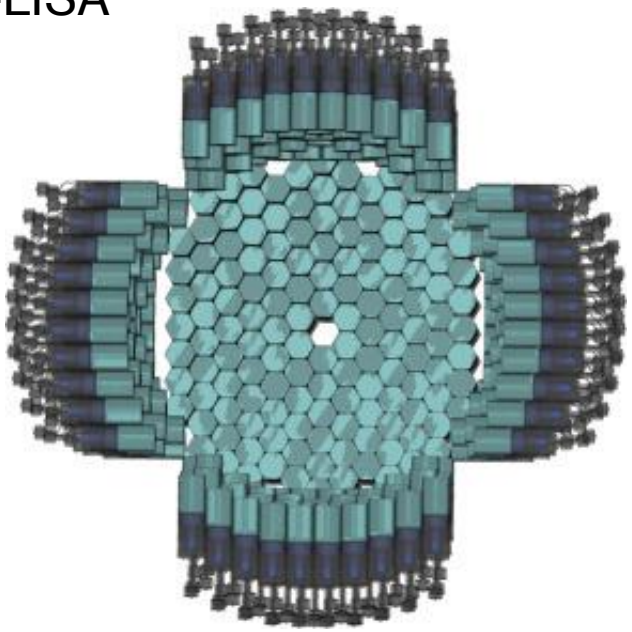


MoNA-LISA

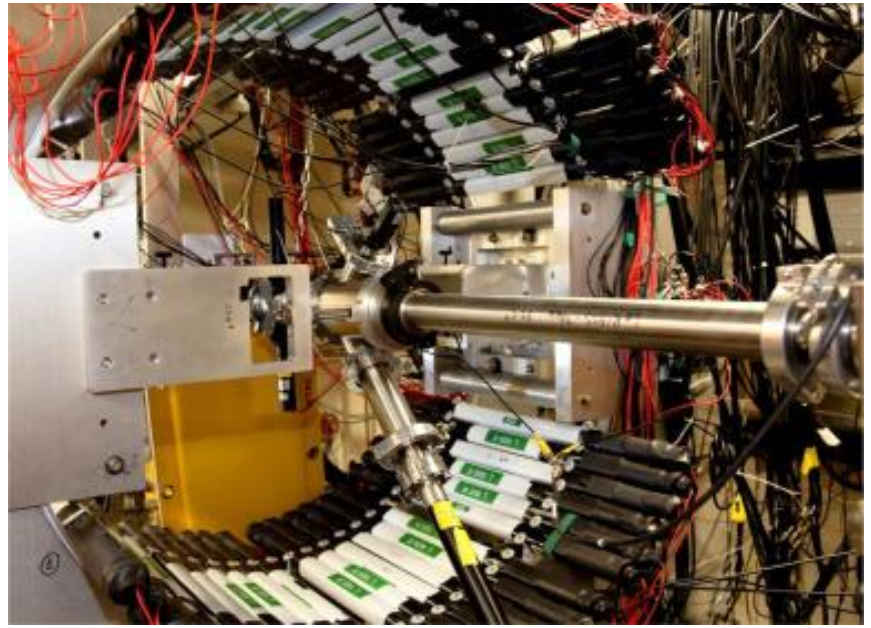
- Typically, scintillator detectors or gas-filled



HeBGB



NEDA



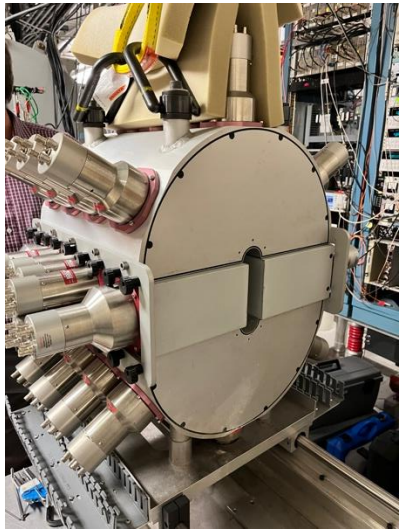
VANDLE

Gamma-Ray Detection

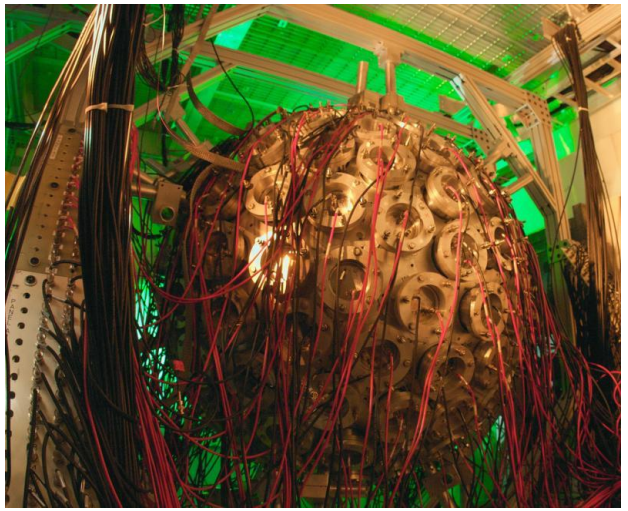
- Several types: semi-conductors (HPGe), scintillators (Nal, LaBr, CeBr, ...)



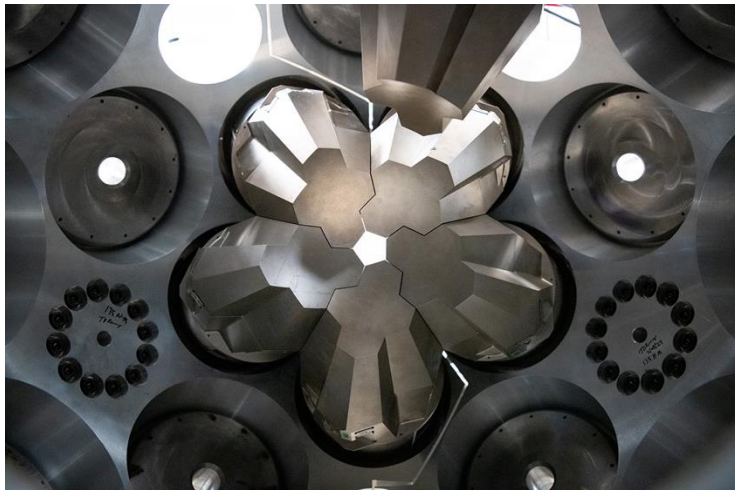
AGATA



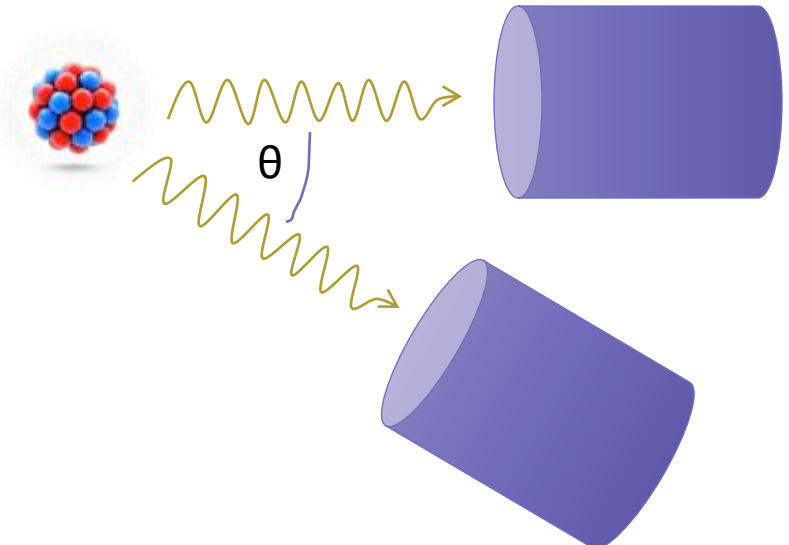
SuN++



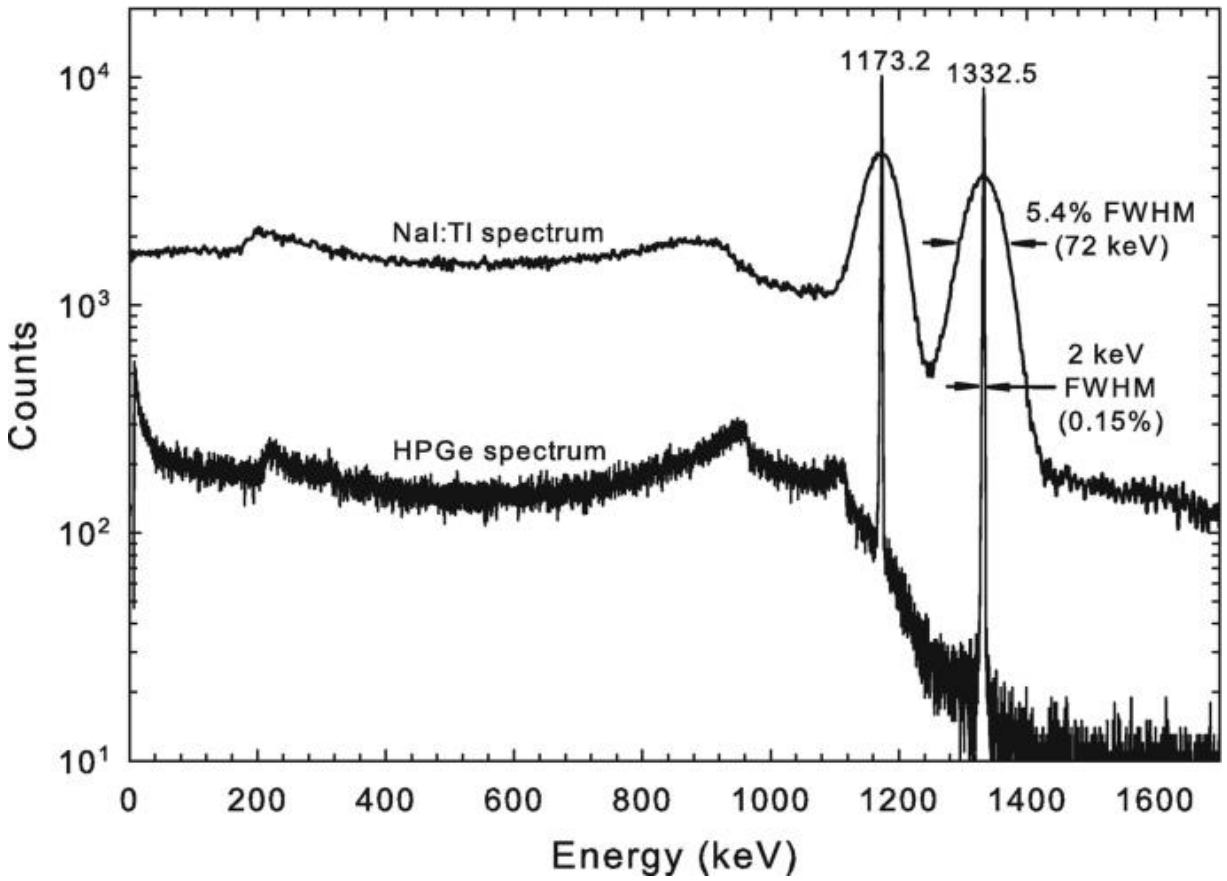
DANCE

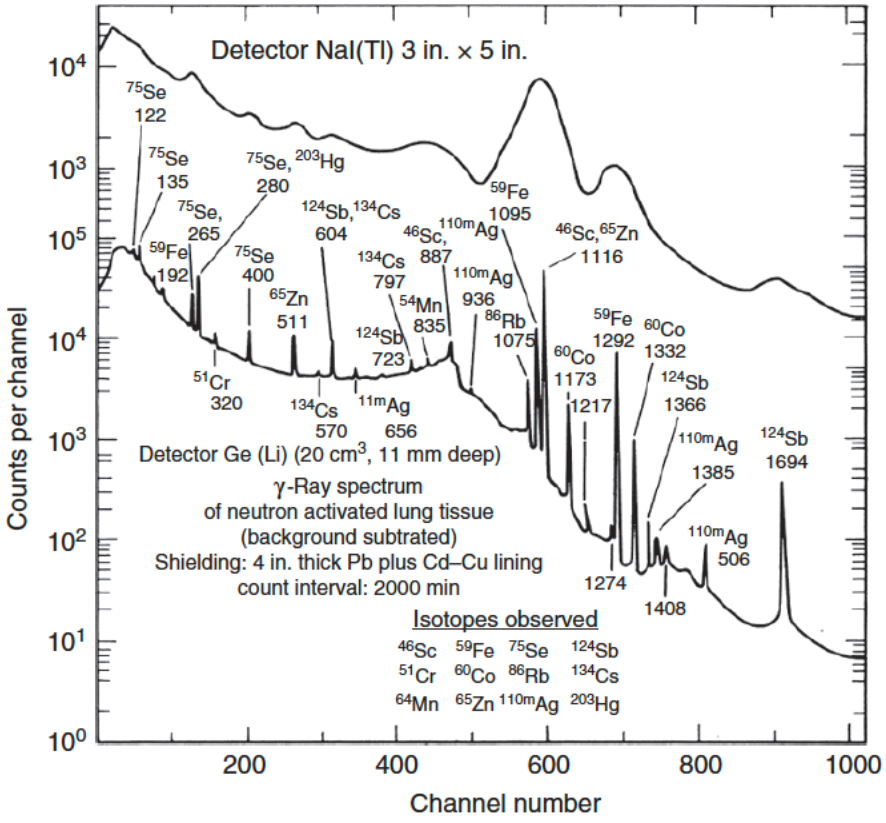


GRETA

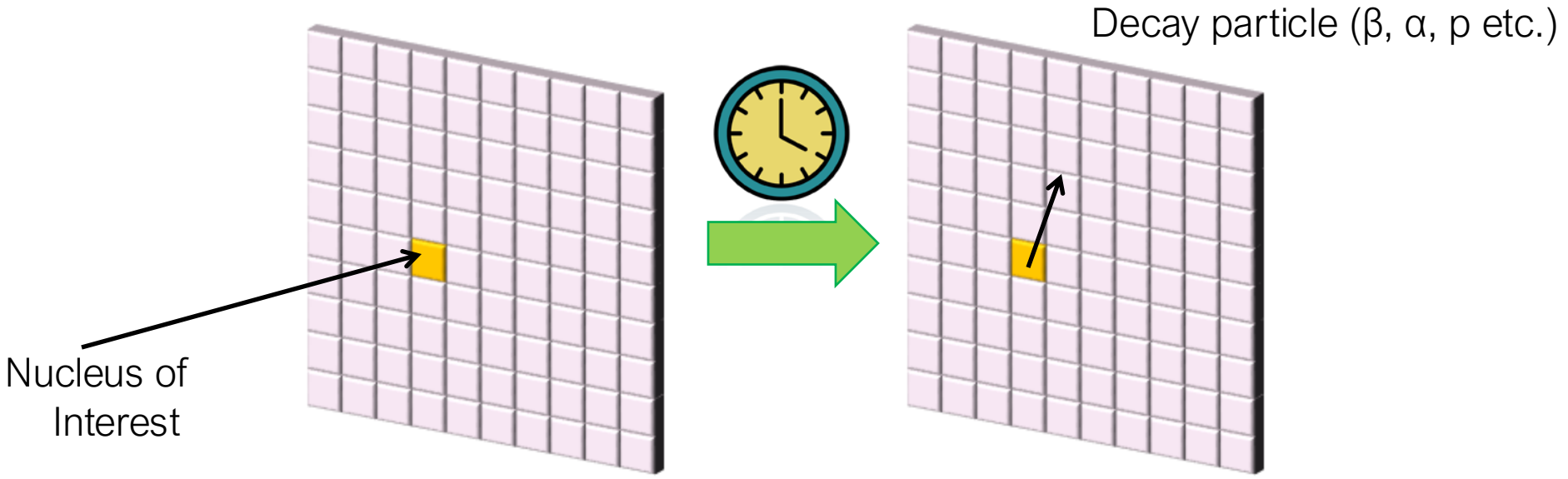


Gamma-Ray Arrays



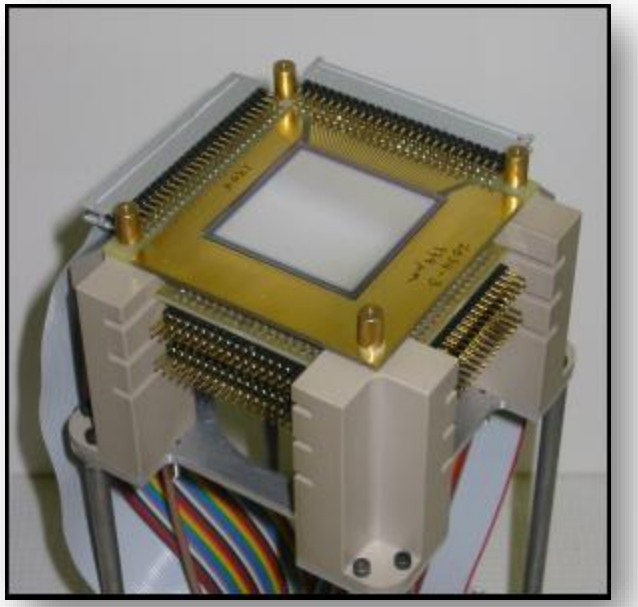


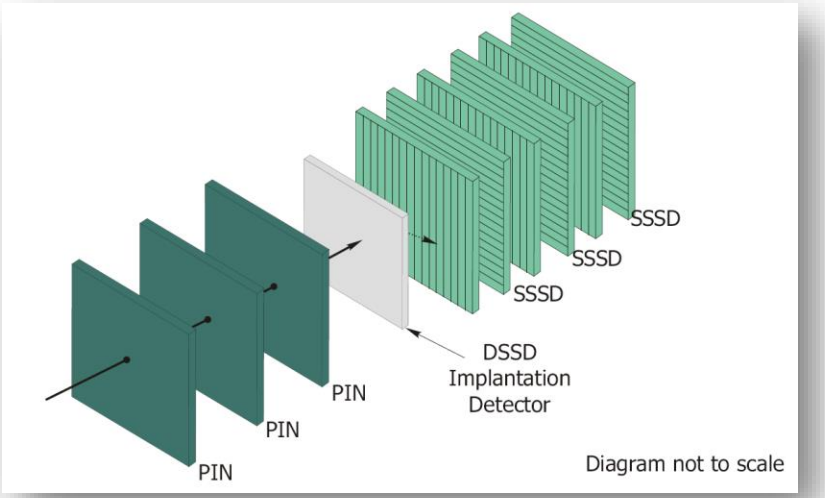
Decay



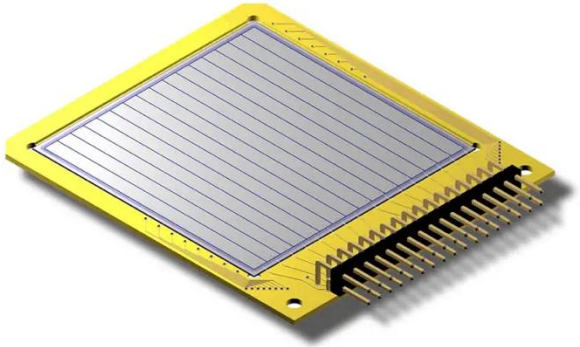
- How do you detect a decay event?
- There are several ways, but the most common is the use of dual gain semi-conductor detectors (usually Si). Observe beam implantation followed by decay corresponding to the half-life of the parent.

Implantation β-decay spectroscopy arrays

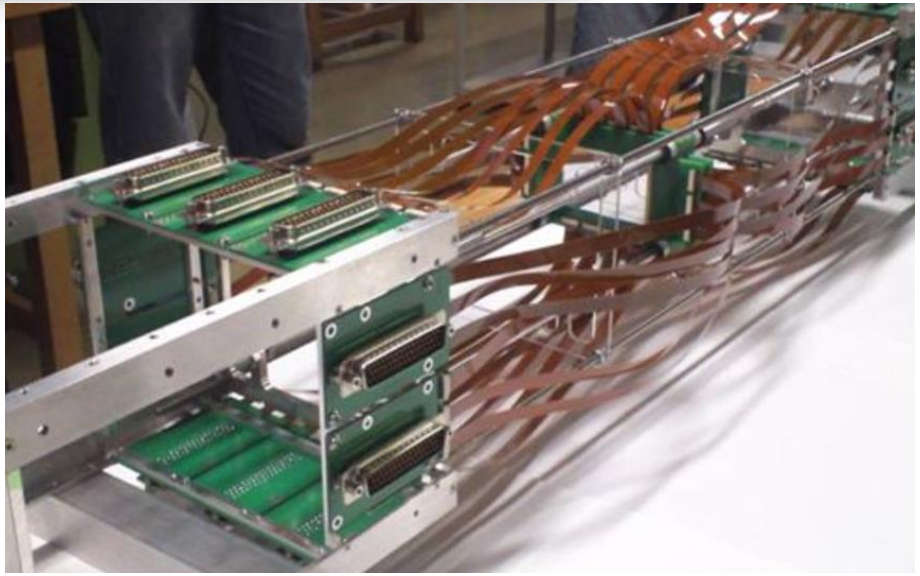




BCS at
NSCL



WAS3ABI for EURICA
at RIBF (RIKEN)

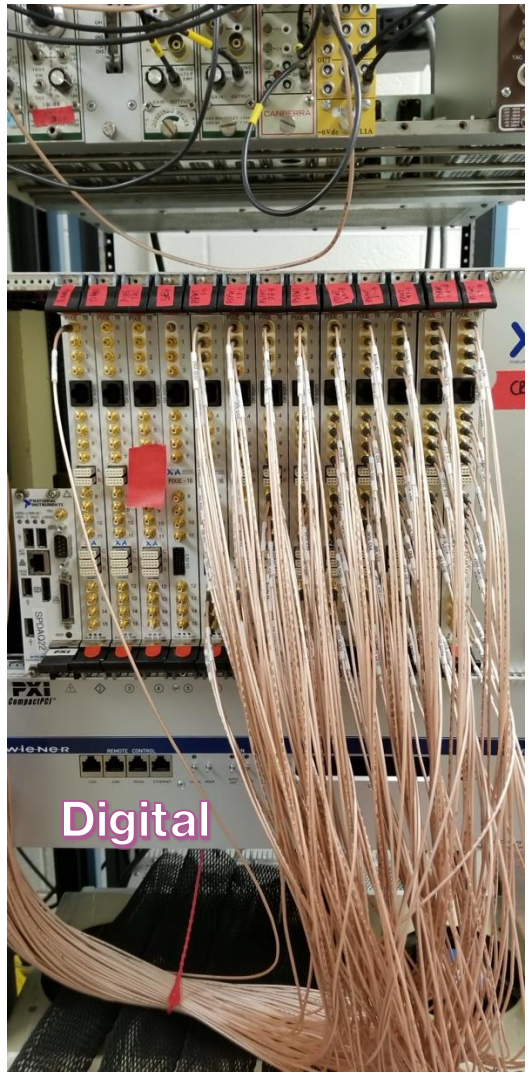


Data Acquisition and Electronics

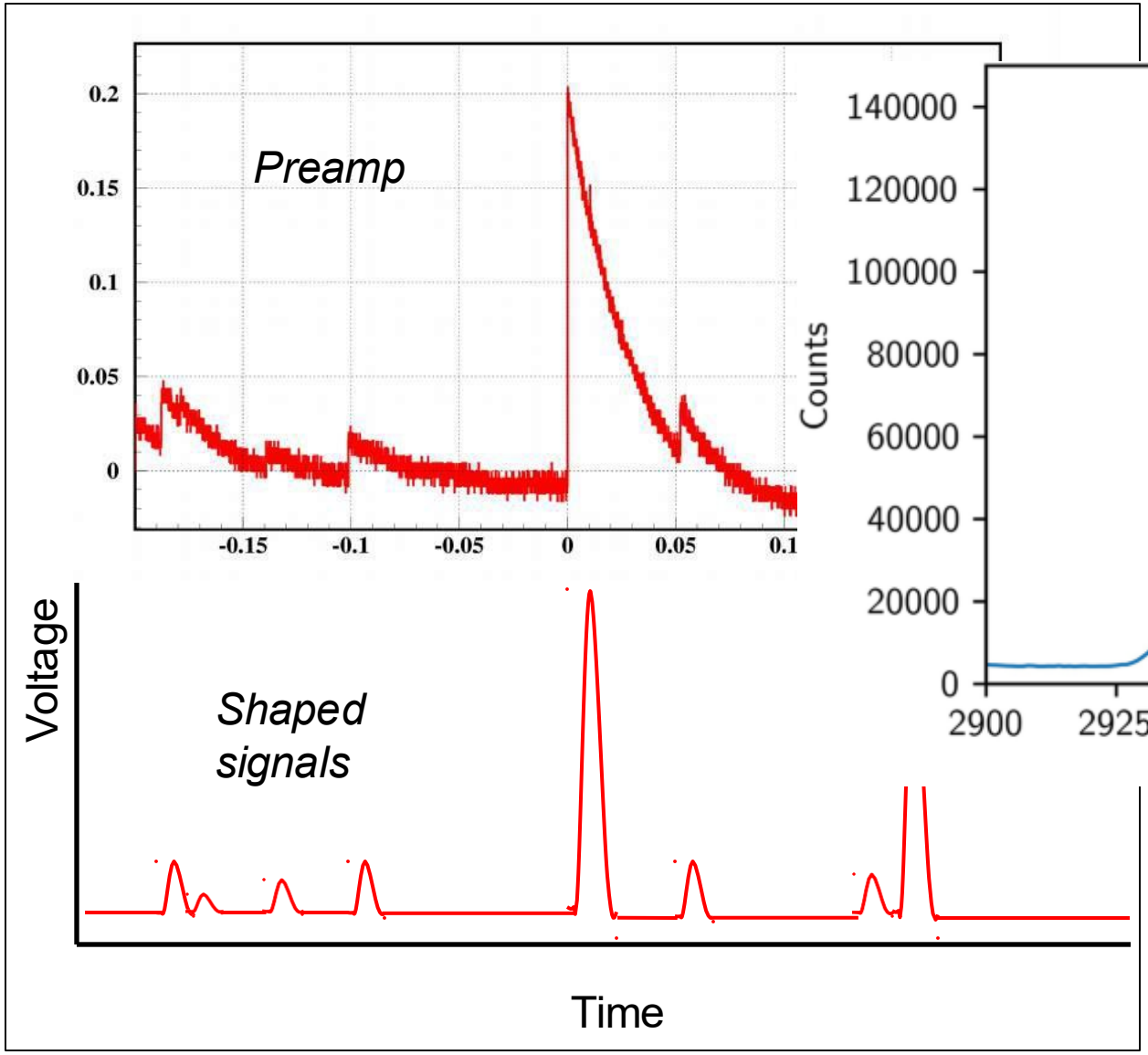
- How do we get signals from the detector to the DAQ?
- This differs slightly based on whether we are using an analog or digital data acquisition system – not because the basic premise differs, but just the order of operations:
  - analog DAQ: signal → shaping → discrimination → logic → event recording
  - digital DAQ: signal → some filtering → event recording → more filtering → logic

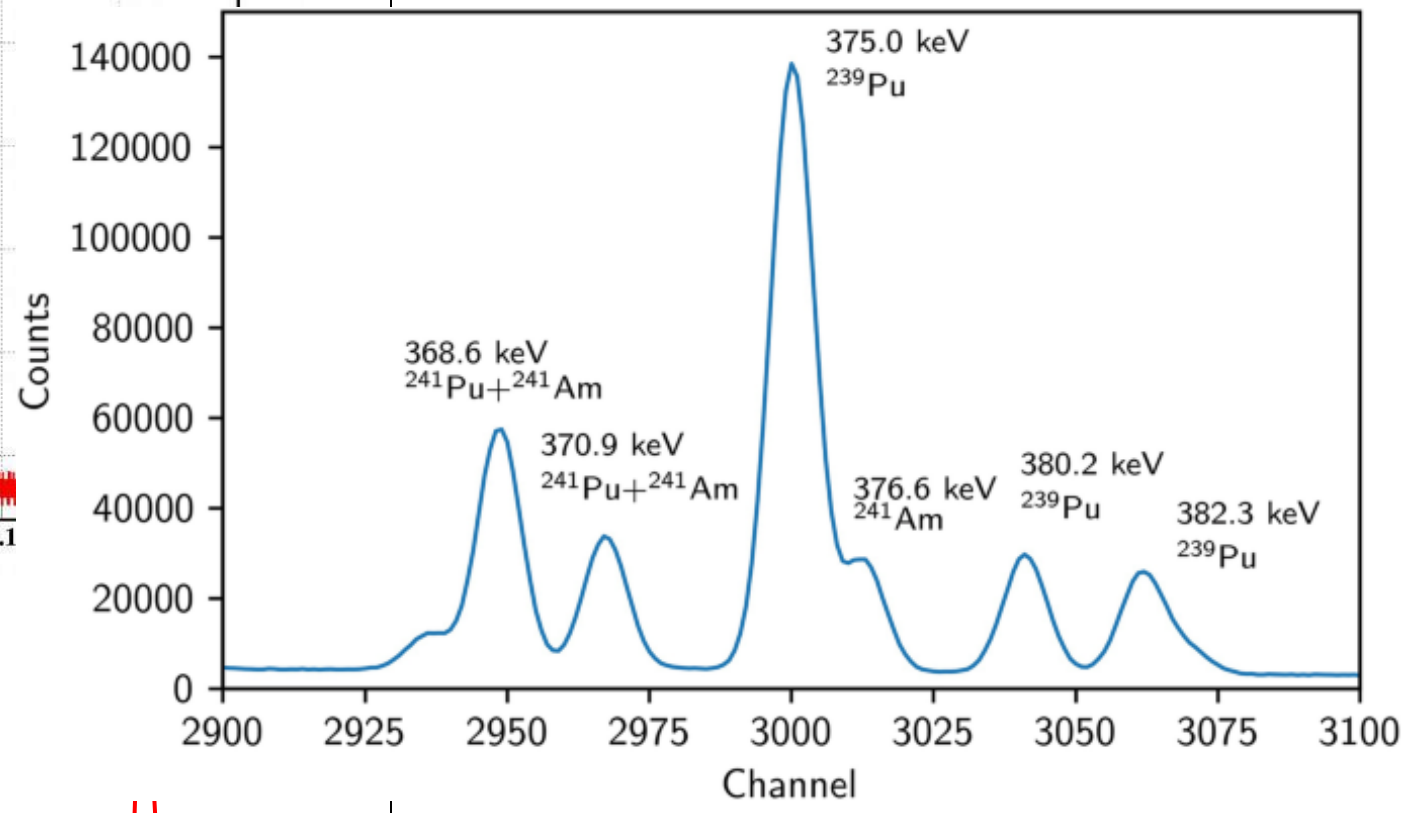
Signals can be processed by standard NIM (Nuclear Instrumentation Modules) or VME (versa module Europa) electronics (for example)





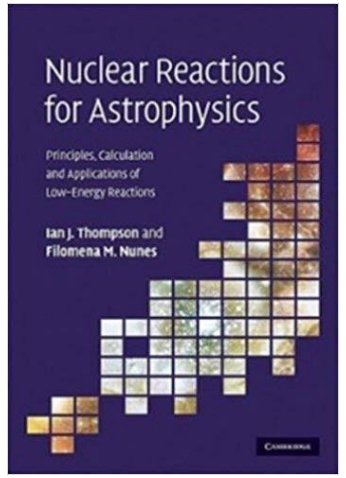
Data Acquisition and Electronics



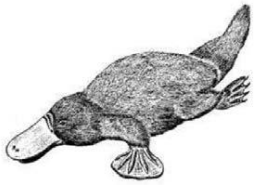


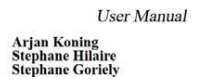
Theory

- Direct Reactions
  - FRESCO (Ian Thompson, LLNL)
    - BFRESCO (BAND collaboration, Bayesian)
    - SFRESCO (Chi-Square searches added)
  - TWOFNR
- Compound Reactions
  - TALYS or EMPIRE
- Codes for decay vary a bit depending on what you want to calculate
  - β-decay properties can be calculated using QRPA (Quasiparticle Random Phase Approximation)











https://fribtheoryalliance.org/content/Resources/codes.php?title=Computer%20codes

Experimental Design

- Get into groups of 3 (include one LE experimentalist)
- Consider how you would measure the following reactions and answer these questions.
- What facility would you use? What detection system? What theory?
  - **Group 1:** $^{16}O(\alpha,n)$
  - **Group 2:** $^{7}Be(p,n)$
  - **Group 3:** $^{99}Mo(n,\gamma)$
  - **Group 4:** $^{82}Se(d,p)$
  - **Group 5**: ^{181}Hf Decay

Wrapping up!



Office Hours Friday 10AM, Edwards 204



Next class: Nuclear Medicine



Paper Outline of Final Due TODAY

Slide Drafts Due 11/25



Questions?

Final Exam Topics

- You should pick a topic from the following list (or propose something to me*). It cannot be closely related to your research. Topics must be approved by me.

- **Islands of Inversion**
- **Shape Coexistence -- Aashish**
- **Non-axial deformation**
- **The N=126 Magic Number**
- **Superheavy Nuclei – Noah (Z=126)**
- **Nuclear Fusion – Muhammad**
- **Reactor Neutrino Anomaly**
- **Neutrinoless double beta-decay -- Joel**
- **Beta-Delayed Particle Emission -- José**
- **s-process nucleosynthesis -- Amadie**
- **r-process nucleosynthesis – Ben**
- **rp-process nucleosynthesis**
- **Nuclear Pasta in Neutron Stars -- Nick**
- **Halo Nuclei -- Mark**
- **Nuclei and Medical Imaging -- Alan**
- **Radiation Therapy Techniques -- Nohora**
- **Medical Isotope Production -- Unish**
- **Effective Field Theory**
- **Rare Isotope Harvesting -- Cade**
- **Neutron Imaging -- Sushma**
- **Astromers**
- **Nuclear Masses and Astrophysics**
- **Nuclear Materials Analysis**
- **Density Functional Theories-**
- **Imaging Shapes of Nuclei -- Parker**
- **Alpha Clustering in Nuclei**
- **Nuclear Forensics* -- Eric**

Due Dates:
10/31 – Topic and at least 5 resources
11/18 – Outline of paper
11/25 – First Draft of Slides

Presentation Dates Volunteers

Dec. 4 2-4PM	Dec. 9 12:20 – 2:20PM
Unish	Joel
Mark	Muhammad
Noah	Amadie
Nohora	Sushma
Aashish	Nick
Cade	José
Eric	Ben
Alan	Parker